

國立中央大學

光電科學與工程學系

博士論文

The investigation and development of the
electro-optic quasi-phase matching photonic devices
in lithium niobate

研 究 生：張瑞文

指導教授：陳彥宏博士

中華民國 103 年 6 月



國立中央大學圖書館

碩博士論文電子檔授權書

(101 年 9 月最新修正版)

本授權書授權本人撰寫之碩/博士學位論文全文電子檔(不包含紙本、詳備註 1 說明)，在「國立中央大學圖書館博碩士論文系統」。(以下請擇一勾選)

(☒)同意 (立即開放)

(☐)同意 (請於西元 _____年____月____日開放)

(☐)不同意，原因是：_____

在國家圖書館「臺灣博碩士論文知識加值系統」

(☒)同意 (立即開放)

(☐)同意 (請於西元 _____年____月____日開放)

(☐)不同意，原因是：_____

以非專屬、無償授權國立中央大學、台灣聯合大學系統圖書館與國家圖書館，基於推動「資源共享、互惠合作」之理念，於回饋社會與學術研究之目的，得不限地域、時間與次數，以紙本、微縮、光碟及其它各種方法將上列論文收錄、重製、與利用，並得將數位化之上列論文與論文電子檔以上載網路方式，提供讀者基於個人非營利性質之線上檢索、閱覽、下載或列印。

研究生簽名：_____張瑞文_____學號：_____952406026_____

論文名稱：The investigation and development of the electro-optic quasi-phase matching photonic devices in lithium niobate

指導教授姓名：_____陳彥宏_____

系所：_____光電_____所 ☒博士班 ☐碩士班

備註：

1. 本授權書之授權範圍僅限**電子檔**，紙本論文部分依著作權法第 15 條第 3 款之規定，採推定原則即預設同意圖書館得公開上架閱覽，如您有申請專利或投稿等考量，不同意紙本上架陳列，須另行加填申請書，詳細說明與紙本申請書下載請至本館數位博碩論文網頁。
2. 本授權書請填寫並**親筆**簽名後，裝訂於各紙本論文封面後之次頁（全文電子檔內之授權書簽名，可用電腦打字代替）。
3. 讀者基於個人非營利性質之線上檢索、閱覽、下載或列印上列論文，應遵守著作權法規定。

國立中央大學博士班研究生

論文指導教授推薦書

光電 學系/研究所 張瑞文 研究生所提之論

The investigation and development of the electro-optic quasi-phase matching
文 photonic devices in lithium niobate

鋇酸鋰電光準相位匹配元件之探討與研發
係由本人指導撰述，同意提付審查。

指導教授 陳長宏 (簽章)

103 年 6 月 16 日

國立中央大學博士班研究生

論文口試委員審定書

光電材料與工程 學系/研究所 張瑞文 研究生

所提之論文

鋇酸鋰電光準相位匹配元件之探討與研發

The investigation and development of the electro-optic quasi-phase matching photonic devices in lithium niobate

經本委員會審議，認定符合博士資格標準。

學位考試委員會召集人

委

員

張明達

陳長宏

楊尚達

游漢輝

鄭登祥

陳弘元

林碩泰

劉秉平

中華民國 103 年 7 月 8 日

Abstract

Under the influence of the electric field in the direction of crystallographic z axis of the periodically poled lithium niobate (PPLN), a periodically varying permittivity along the crystallographic x axis will be produced. Such a device is called EO PPLN grating which can cause the reciprocal vector to compensate the phase mismatch between the incident extraordinary beam and its first order Bragg diffracted extraordinary beam in the device. In this dissertation, we developed a theoretical model to investigate and analyze the Bragg diffraction behavior of a Gaussian beam that interacts with such a device. This model can also be used to evaluate the performance of volume gratings in an optical system.

For an electric field along the crystallographic y axis of the PPLN, the polarization state of the light which propagates along x axis in the PPLN can be modulated efficiently, and it has been explored and exploited to demonstrate efficient and fast polarization-mode converter (PMC). In this dissertation, we integrated two PPLN EO PMCs with a PPLN optical parametric gain medium in a monolithic LiNbO_3 to achieve unique spectral manipulations in an optical parametric oscillator (OPO) system. With this kind of EO effect in, say an, annealed proton-exchanged PPLN waveguides, we report the design and experimental demonstration of electro-optically active TM-guided to TE-radiation mode converters.

The devices demonstrated in this research have the potential for practical application. To further integrate the devices with other PPLN or APPLN structure with different optical functions in a monolithic LiNbO_3 crystal, such devices can find many applications in optical communications or an electro-optic Q-switched and wavelength-tunable laser system.

摘要

當外加電場沿著本研究中第一種週期性極化翻轉鈮酸鋰的晶軸 z 方向，此時在晶軸 x 方向會產生週期性的介電常數變化，這種元件被稱為週期性極化翻轉鈮酸鋰電光光柵，它的倒晶格向量可用來補償在光柵中的 0 皆非尋常入射光和 1 皆非尋常繞射光的波向量不匹配。我們在本文中發展一種理論模型，用來分析並且觀察一個有限腰寬的高斯光束；當它沿 1 皆布拉格角度入射週期性極化翻轉鈮酸鋰電光光柵時的布拉格繞射行為。這種理論模型可以根據入射光束的束腰，來評估體積光柵在光學系統中的表現。

當外加電場沿著我們第二種週期性極化翻轉鈮酸鋰的晶軸 y 方向，此時沿 x 軸方向傳播光的偏振方向可以被快速的調製，這種元件我們稱為偏振模態轉換器。在本文中我們整合兩個偏振模態轉換器與一個週期性極化翻轉鈮酸鋰光參量增益介質在同一個鈮酸鋰晶體上，用它來產生並且快速調製光參量震盪器的輸出頻譜。第三種元件是將退火式質子交換波導製作在週期性極化翻轉鈮酸鋰晶體上，當外加電場沿著週期性極化翻轉鈮酸鋰的晶軸 y 方向，我們設計並且展示了一種有效率的主動式電光模態轉換器，它可以被用來轉換波導中的傳播模態與輻射模態。

本研究中所展示元件具有實際應用的潛力，當將它們與具有不同光學功能的週期性極化翻轉鈮酸鋰與非週期性極化翻轉鈮酸鋰結構結合在一塊鈮酸鋰晶體上，這些元件可在光纖通訊與電光可調式脈衝雷射系統上具有更多種的應用。

Acknowledgement

感謝我的父親、母親，若不是您們的支持我是無法完成學業的。我的老婆、小龍喵、家人以及游漢輝老師，謝謝您們的幫助與體諒，沒有這些我是堅持不下去的。

在中央的8年，我遇到了一個教育家；游漢輝老師。游老師引我進入學術的大堂；並教我規矩，當他的助教我收穫許多，尤其是在他退休後，花了更多時間與精神在教學上。他在課堂上對於基礎理論的分析，常讓我眼睛為之一亮；並且得到許多靈感，但是在目前的環境下，他的理念似乎不受歡迎，他很重視作業，認為作業除了是練習課堂所學的之外；更是練習溝通與表達的訓練，所以他總是寫了一堆評語指出不合邏輯與表達錯誤之處，他不喜歡考考古題，經過我的觀察，這是受填鴨式教育學生最討厭的地方，就像煮飯；給了你固定的食材，要你烹煮某樣菜，與自己去市場挑選食材的差別，研究所了、該自己買菜了吧。游老師的為人樸實體諒人，而這些的果報，就是畢業的學生不時地回來探望他關心他，不要被游老師上課的嚴肅一面嚇到，私底下他很不一樣的。我的指導老師陳彥宏老師，謝謝他引我進入非線性光學的領域，也謝謝他尊重我的意願讓我自己選擇想做的東西。

在這8年，我看了很多意想不到的事，雖然很多事是不好的，但是讓我更了解不同的人與人性，多一些經驗與經歷也不錯。我很喜歡學弟問我問題，並且經過思考後質疑我的說法，在學弟不斷的反饋與溝通中，糾正了我不少理論上的錯誤觀念，而這方面我印象深刻的學弟有文豪，當博班的學長，千萬不要一副我是對的樣子，當你懂得越多，你越瞭解無法面面俱到，你會越謙虛的，還有博班也有教育學弟的一部份責任，不要把實驗室搞得像公司一樣了。我很欣賞光緒堅持思考後再做實驗的態度，他不畏畢業的壓力；發現了元件設計上的錯誤，大家是來實驗室學習獨立思考解決問題的能力，不是來當科技秘書或操作員的，教育是要極大的耐心與溝通的。另外我對錦峯的韌性感到驚訝，對耀賢的不畏感到欽佩，有太多值得我學習與警惕的，謝謝您們。在做實驗的過程中，我更了解為何期刊論文和被引用數比期刊本身的好壞來的重要，因為很多研究都是無法重現或有重大問題的，我相信這當中有許多無意的過失，但我也相信有不少是刻意的編造，聽過太多東方與西方的差別，人家大部分都按部就班一步一步地來，求真、求實，我們大部分的研究求快、求好、求發到好的期刊，業界的短視近利與困境學界也不遑多讓。我常在想人總會一死，死的時候自己面對自己，什麼名與利都跟你無關了，若這樣、我還是會選擇目前的生活理念嗎？在這8年中我是對得起自己的。

Contents

Abstract	I
摘要	II
Acknowledgement	III
List of Figures	V
Chapter 1 Introduction	1
Chapter 2 Finite beam Bragg diffraction in periodically poled lithium niobate electro-optic grating	
2.1 Theoretical development.....	13
2.2 Other EO induced effects.....	17
2.3 Device and experimental setup.....	21
2.4 Experimental results and theoretical simulation analysis.....	23
2.5 Chapter summary.....	34
Chapter 3 An optical parametric oscillator using periodically poled lithium niobate electro-optic polarization-mod converter	
3.1 Device.....	36
3.2 Theoretical analysis.....	37
3.3 Experimental setup.....	41
3.4 Experimental Results.....	42
3.5 Chapter summary.....	45
Chapter 4 Electro-optic guided-to-radiation mode conversion in annealed proton-exchanged periodically poled lithium niobate waveguides	
4.1 Annealed proton-exchanged LiNbO ₃ waveguide.....	46
4.2 Device design.....	48
4.3 Device fabrication.....	52
4.4 Performance characterization.....	54
4.5 Chapter summary.....	62
Chapter 5 Conclusion and further study	63
Reference	66

List of Figures

Figure 1-1. Schematic of the periodically poled lithium niobate.....	2
Figure 1-2. Scheme of an electric field along the z axis of LiNbO ₃	2
Figure 1-3. Scheme of a PPLN EO grating for the electric field along the crystallographic z axis.....	3
Figure 1-4. Scheme of an electric field along the y axis of LiNbO ₃	6
Figure 1-5. Energy conversion between the extraordinary wave and the ordinary wave when (a) the QPM condition is satisfied and (b) the QPM condition is not satisfied.....	7
Figure 1-6. Schematic diagrams show the linewidth control and spectral tuning of OPOs.....	8
Figure 1-7. Effective index distributions scheme of TM and TE modes of our APE LiNbO ₃ Waveguide.....	10
Figure 1-8. Scheme of a PPLN EO GTR PMC.....	11
Figure 1-9. Electrode configurations of EO GTR PMCs.....	11
Figure 2-1. Schematic of a PPLN EO Bragg grating used as an illustration for our theoretical development.....	13
Figure 2-2. Development of a beam propagating in a high-refractive-index domain layer and passing through the dielectric interface between the layer and its neighbor low-refractive-index layer in a PPLN EO grating when the incident angle (a) $\theta_{B1} > \theta_c(V)$ or (b) $\theta_{B1} < \theta_c(V)$. (c) (Calculated) specific incident angle ($\theta_c(V)$) of the beam that leads to its refracted beam incident at the critical angle $\theta_c(V)$ of the dielectric interface as a function of the applied voltage.....	18
Figure 2-3. Calculated penetrating distance q as a function of the applied voltage.....	20
Figure 2-4. (a) Schematic configuration of a PPLN EO grating device. (b) A typical microscopic image of the HF-etched +z surface of the fabricated PPLN crystal.....	21
Figure 2-5. The schematic arrangement of the experimental setup.....	22
Figure 2-6. Measured (black and green curves) and calculated (dashed red and blue curves for square-wave and cosinusoidal-wave shape gratings, respectively) 0 th order transmittances of a 1064-nm laser after traversing the PPLN EO Bragg device as a function of the applied voltage for beam waist	V

radii of (a) 28 and (b) 135 μm . (c) the same measurement results (black and green curves) as plotted in (a) and (b) but with simulations (dashed curves) based on a modified Fourier amplitude of the modulated permittivity (see text). (d) A periodic trapezoidal grating form..... 23

Figure 2-7. (a) transmittance of the 0th order harmonic wave as a function of the driving voltage of the PPLN EO Bragg device for plane wave elements of different absolute values of the dephasing factor D. The right Figures show the angular spectra and the corresponding dephasing factor distribution of the incident Gaussian beams with waist radii of (b) 28 and (c) 135 μm 25

Figure 2-8. Measured (black curves) and calculated (red curves) beam intensity distribution along the central horizontal line of the beam profiler (corresponding to the dashed lines in the insets) for various specific operating voltages of the device for incident Gaussian beam waist radii of 28 μm (left Figures) and 135 μm (right Figures). The insets show the captured images of the corresponding 2D intensity distributions..... 29

Figure 2-9. Calculated output angular spectra of two Gaussian beams of waist radii of 28 μm (left Figures) and 135 μm (right Figures) after traversing the PPLN EO Bragg device under those specific operating voltages used in figure 2-8. The vertical scales are directly comparable for those results obtained from the same incident beam size..... 31

Figure 3-1. Schematic arrangement of the 1064 nm pumped OPO system at the 1532 nm band based on a monolithic LiNbO₃ integrating a 3 cm long, 29.8 μm period PPLN OPGM and two 1.25 cm long, 21 μm period PPLN EO PMCs..... 37

Figure 3-2. A scheme of the buildup process of the OPO signal..... 39

Figure 3-3. Calculated e-polarized wave spectra after the first round trip in the device shown in Figure 3-2 at $E_y = 93 \text{ V/mm}$ (dotted line) and 1168 V/mm (solid black line), provided a flat-spectrum signal wave is generated from the OPGM..... 40

Figure 3-4. A schematic drawing of the experimental setup..... 42

Figure 3-5. (Color on line) Measured (solid black line; blue/red on line) and calculated (dashed gray line) output signal spectra from the OPO system at $T_{\text{OPGM}} = 45 \text{ }^\circ\text{C}$, $T_{\text{PMC1}} = T_{\text{PMC2}} = 44.3 \text{ }^\circ\text{C}$, and at $(E_{\text{PMC1}},$

E_{PMC2})=(a) (1290, 980) V/mm; (b) (1060,700) V/mm (i) and (900, 800) V/mm (ii); and (c) (0, 600) V/mm..... 43

Figure 3-6. (Color on line) Measured (solid black line; blue on line) and calculated (dashed gray line) output signal spectra from the OPO system at $T_{OPGM}=42$ °C, $T_{PMC1} = T_{PMC2} = 42.8$ °C, and at (E_{PMC1} , E_{PMC2}) =(a) (940, 1100) V/mm and (b) (1040, 1050) V/mm..... 44

Figure 4-1. Scheme of the extraordinary refractive index distribution $n(z)$ along the central vertical dashed line of our APE waveguide..... 48

Figure 4-2. Calculated normalized overlap integral as a function of the effective refractive index and the corresponding escaping angle of the radiation mode for the coupling between the TM_{00} and TE radiation modes in an APE LiNbO3 waveguide at $\lambda_0 = 1520$ nm, at 40°C..... 51

Figure 4-3. Microscopic image of a portion of the + z surface of the fabricated APE PPLN EO waveguide device..... 53

Figure 4-4. Experimental setup for the performance characterization of the APE PPLN EO waveguides. 54

Figure 4-5. Measured spectral transmittances (T_{EO}) of the TM mode from a 24- μ m-period APE PPLN waveguide for various EO voltages at 35°C..... 55

Figure 4-6. (a) is the schematic pattern of the planar waveguide which is used for estimation of the broad conversion bandwidth. (b) shows the calculated normalized overlap integral as a function of the effective refractive index for the coupling between the TM_{00} and TE radiation modes in the planar waveguide at $\lambda_0 = 1520$ nm at 40°C..... 57

Figure 4-7. It shows the mode overlap integral as a function of the effective refractive index for the coupling between the TM_{00} and TE radiation modes in the planar waveguide at $\lambda_0 = 1500$ nm and 1580 nm at 40°C..... 58

Figure 4-8. (a) Measured spectral transmittances (T_{EO}) of the TM mode from the APE PPLN EO waveguides of various grating periods (not shown for $\Lambda = 10$ and 50 μ m) when operated at 250 V and 35°C. (b) Mode conversion efficiency (η_c) of the device as a function of the PPLN period at 1520-nm

wavelength.....	59
Figure 4-9. The schematic pattern of the applied electric field distribution.....	60
Figure 4-10. Far-field intensity profiles of the 1550-nm TM_{00} guided mode measured from an APE waveguide in the single-domain section with an EO voltage of 0 V (left) and 250 V (right).....	61
Figure 4-11. Far-field intensity profiles of the 1520-nm TM_{00} guided mode measured from an APE waveguide of grating period $24\mu\text{m}$ with an EO voltage of 0 V (left) and 250 V (right).....	61
Figure 5-1. A schematic arrangement of the 1064 nm pumped OPO system using an APPLN.....	65
Figure 5-2. A schematic arrangement of the pulsed IOPO constructed in a diode-pumped 1064-nm Nd:YVO ₄ laser using the device as simultaneously the laser Q-switch, the electro-optically spectral tuning, and the OPG.....	65

Chapter 1

Introduction

Lithium niobate (LiNbO_3) have emerged as an important material due to its useful optical phenomena including; electro-optics, piezoelectricity, high dielectric constants, optical waveguiding, frequency generation, etc. Moreover, it has the large electro-optic and non-linear coefficient, which can be exploited to produce novel devices based on LiNbO_3 . LiNbO_3 can be periodically poled for a given grating period Λ . In such a structure, the direction of the spontaneous polarization of the LiNbO_3 is inverted periodically as a function of position. As shown in Figure 1-1, the nonlinear coefficient d_{33} caused by this periodic domain structure is as a function of x , and it can be expressed by

$$d(x) = d_{33} f(x), \quad (1.1)$$

where $f(x)$ denotes the direction of the spontaneous polarization of the LiNbO_3 . $f(x)$ takes a value of 1 or -1 for the x in the positive domains or negative domains, respectively. It is useful to express $d(x)$ in terms of infinite Fourier series as

$$d(x) = d_{33} \sum_{m=-\infty}^{\infty} G_m \exp(-iK_m x), \quad (1.2)$$

where G_m and $K_m = 2\pi m / \Lambda$ are the m -th order Fourier coefficient and grating vector, respectively. The reciprocal vector k_m may compensate the wave-vector mismatch of a wave-energy coupling process in the PPLN and has been widely used in nonlinear frequency conversions [1]. This process is called quasi-phase-matching (QPM) [2], and such materials are called QPM materials.

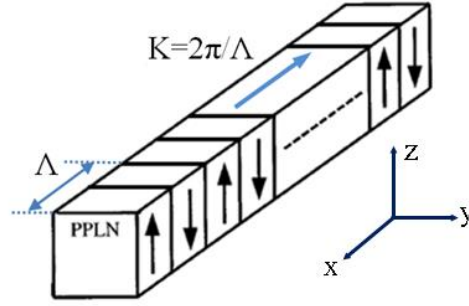


Figure 1-1. Schematic of the periodically poled lithium niobate. x , y , and z represent the crystallographic axes. The arrows in the PPLN indicate the directions of spontaneous polarization.

Compared to the conventional birefringence phase-matching, this technique has many advantages which has led to a rapid advance of the optical parametric oscillator (OPO) technique [1]. In addition to the nonlinear coefficient, the electro-optic (EO) coefficients are also modulated periodically due to the periodic domain structure.

As shown in Figure 1-2, we consider the case when an electric field is applied along the crystallographic z axis of LiNbO_3 .

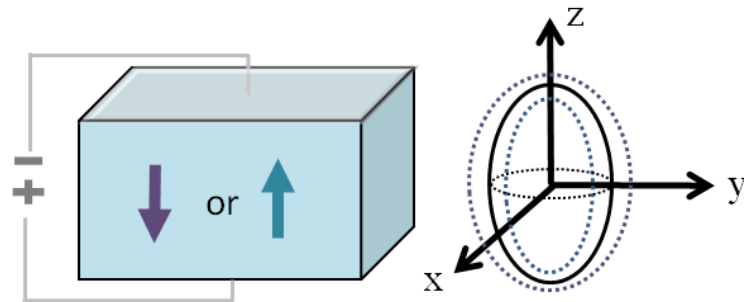


Figure 1-2. Scheme of an electric field along the z axis of LiNbO_3 . x , y , and z represent the crystallographic axes. The arrows in the LiNbO_3 indicate the directions of spontaneous polarization. The solid black shape represents the original index ellipsoid of LiNbO_3 without perturbation. The dotted blue and purple shapes represent the perturbed index ellipsoids of the positive domain and the negative domain, respectively.

Under this condition, LiNbO₃ remains uniaxial with the same principal axes, but the dielectric tensor becomes

$$\varepsilon_{ij} = \varepsilon_0 \begin{pmatrix} n_0^2(1 - n_0^2 r_{13} E_z) & 0 & 0 \\ 0 & n_0^2(1 - n_0^2 r_{13} E_z) & 0 \\ 0 & 0 & n_e^2(1 - n_e^2 r_{33} E_z) \end{pmatrix}. \quad (1.3)$$

And the refractive indices for the wave polarized along z-axis in the x-y plane are

$$\begin{aligned} n_+ &= n_e - \frac{1}{2} n_e^3 r_{33} E \\ n_- &= n_e + \frac{1}{2} n_e^3 r_{33} E, \end{aligned} \quad (1.4)$$

where E is the field amplitude, r_{33} is the EO coefficient, n_e represents the unperturbed refractive index of the extraordinary wave, n_+ and n_- are the perturbed refractive index for the extraordinary wave in the positive domains and the negative domain, respectively. In chapter 2, we demonstrate a PPLN EO grating which uses an EO effect accessing the diagonal dielectric tensor element of the PPLN. Such a device was first investigated by Yamada [3]. In Figure 1-3, it shows the scheme of the device. When driven by an external electric field along the crystallographic z axis, a periodically modulated refraction index along the x axis will be produced in the device according to Eq. (1.3).

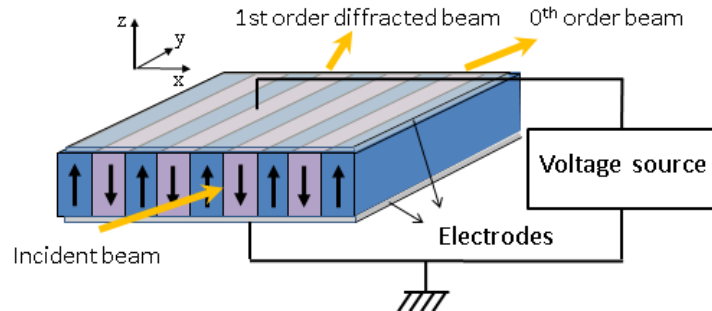


Figure 1-3. Scheme of a PPLN EO grating for the electric field along the crystallographic z axis. x , y and z represent the crystallographic axes. The arrows in the PPLN indicate the directions of spontaneous polarization.

In this device, a reciprocal vector is provided by the periodic modulation of the diagonal dielectric tensor element of the PPLN to compensate for the phase mismatch between the incident extraordinary beam and its first order diffracted extraordinary beam propagating in the x-y plane. Such a device has attracted much attention and found a variety of important applications [3-5]. Advantages of a PPLN EO Bragg cell include having a lower driving voltage and a five-fold reduction in switching time compared with acousto-optic (AO) devices as well as having a suppressed acoustic mode excitation because of the periodically poled structure [6]. The study of the diffraction characteristics of a PPLN EO Bragg grating continued actively over the past decade [3-7]. However, as far as we know, the theoretical studies of this device were dealing with the interaction of it with plane waves of infinite cross section [8] rather than with a finite beam which is the actual arrangement when this kind of device is put in practical use. Accordingly we took up the investigation of the Bragg diffraction of a finite beam in such a device, and we present here the characteristics of the Bragg diffraction of a Gaussian laser beam in such a device, and we found they can exhibit features very different from those predicted by a theory using the plane-wave approximation. In particular, we found the diffraction efficiency and the diffraction mode pattern of a PPLN EO Bragg device are closely related to the waist size of the incident beam and the grating strength. Such a diffraction behavior could be a critical issue to most of the applications using the present device (e.g., when it works as a laser intracavity element [5]). To understand the characteristics of the device, we developed a theoretical model to treat the diffraction behavior of a Gaussian beam interacting with a PPLN EO Bragg device. In contrast to some other models built for a phase grating (such as a photosensitive holographic grating), the present model has benefited from several important features of the PPLN EO Bragg grating. First, PPLN has a definite grating profile and a wide transparent region (350 to 5200 nm), which largely

increases the reliability of the model because problems arising from the non-uniformity of the grating structure and the wave absorption can be ignored. Second, the grating strength of our device is electrically controllable, which allows us to observe and investigate some interesting characteristics of the finite beam diffraction that are dependent on the applied voltage.

In the present treatment, the incident wave vector is set to lie in a plane that contains the grating vector of the PPLN EO Bragg device. Thus the Bragg diffraction here is essentially of a two-dimensional problem. We approached the problem with a model based on the coupled-wave theory [9] along with the decomposition of the incident wave into its component plane waves [10]. The decomposition approach is used because it can help simplify the modeling for the spatially fast-varying beams that interact with the grating and also because it is appropriate to deal with problems involving a uniform grating in a medium with plane-parallel boundaries [11] as is the case in a PPLN. Furthermore, to obtain the coupled-wave equations, we adopted a field representation approach in which the incident Gaussian beam is first expanded into the component plane waves that composes of the original Gaussian beam, and each plane wave element upon entering the grating is expanded into a sum of space-harmonic waves with the magnitude of the wave-vector equal to that of the undiffracted wave-vector [9]. To make clear the Bragg coupling mechanism in such a device, we compare the component plane wave elements before entering the device with the space-harmonic waves after having traversed the device. The present model can also be applied to deal with a Bragg diffraction involving a non-Gaussian beam and a general grating profile. In addition, this model can be readily used to solve the higher-order diffracted beams problems in the case when modulation refraction index in the grating is large.

In contrast to the chapter 2, we consider the case when the electric field is along the crystallographic y axis of LiNbO₃ in chapters 3 and 4, and the dielectric tensor becomes

$$\varepsilon_{ij} \approx \varepsilon_0 \begin{pmatrix} n_o^2 & 0 & 0 \\ 0 & n_o^2 & -n_o^2 n_e^2 r_{51} E_y \\ 0 & -n_o^2 n_e^2 r_{51} E_y & n_e^2 \end{pmatrix}. \quad (1.5)$$

As shown in Figure 1-4, the new index ellipsoid has its new principal axes rotated at an angle θ about the x axis with respect to the unperturbed principal axes at this time.

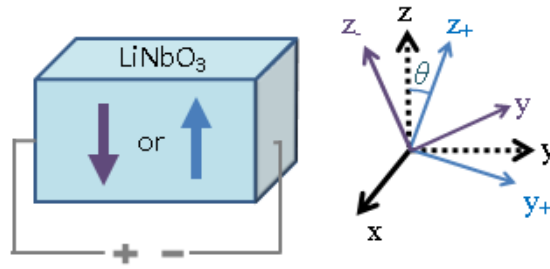


Figure 1-4. x, y and z represent the principal axes of the original index ellipsoid. x₊, y₊, and z₊ are perturbed principal axes for the case of the positive domain and the negative domain, respectively.

This small angle θ is expressed by

$$\theta \approx \frac{r_{51} E_y}{\left(1/n_e^2\right) - \left(1/n_o^2\right)}, \quad (1.6)$$

where E is the field intensity, r_{51} is the involving EO coefficient, n_o and n_e represent the refractive indices of the ordinary wave and extraordinary wave, respectively. At this time, the coupling between the extraordinary wave and ordinary wave of the unperturbed principal axes is established. If an electric field is applied along the y axis of PPLN, such an EO effect accessing the off-diagonal dielectric tensor element of the

PPLN has been explored and exploited to demonstrate polarization-mode converter (PMC) as a Šolc-filter-like polarization rotator [12] and narrowband wavelength filters [12, 13]. Under the QPM condition, the polarization of light which propagates along x axis in the PPLN could be modulated efficiently and fast by the applied electric field along the y axis. In Figure 1-5, it shows the conversion process between the ordinary and extraordinary waves under the EO effect for a sample with the domain thickness of $10.31\text{ }\mu\text{m}$ and the period of 500, the first-order QPM condition corresponds to the wavelength of 1550 nm [12].

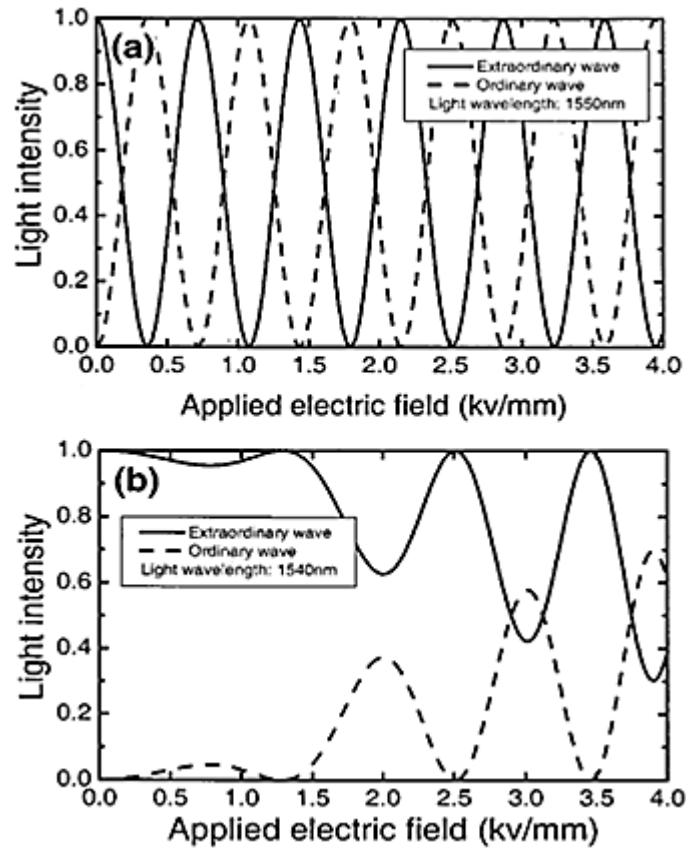


Figure 1-5. Energy conversion between the extraordinary wave and the ordinary wave when (a) the QPM condition is satisfied and (b) the QPM condition is not satisfied. The solid line and the dashed line correspond to the intensity-varying curve of the two waves along the propagation direction, respectively [12].

Optical parametric oscillators (OPOs) are popular sources for producing wavelength-tunable coherent radiation for various applications. Besides the tunability, many applications such as spectroscopy, communication, LIDAR, and remote sensing demand light sources working at a narrow linewidth. Thus, a spectral narrowing mechanism needs to be introduced to the OPO (especially for pulsed OPOs characterized by a broader bandwidth) to meet the requirements of these applications. As shown schematically in Figure 1-6, two methods have been studied to achieve that purpose, among which the insertion of wavelength-selective elements such as etalons or grazing-incidence gratings in the cavity have been the most common practice [14].

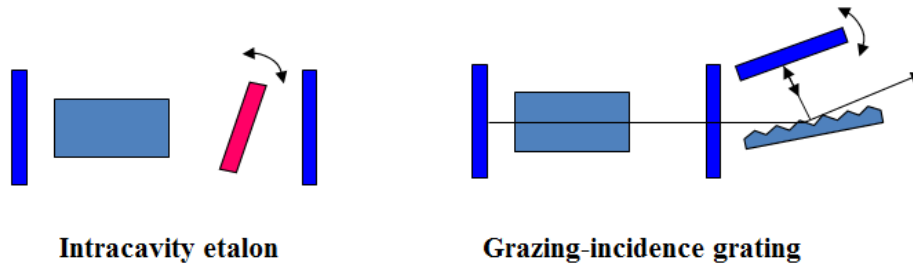


Figure 1-6. Schematic diagrams show the linewidth control and spectral tuning of OPOs.

Nevertheless, the addition of intracavity elements inevitably increases the systems loss as well as the systems complexity. The recent development of glass-based volume Bragg gratings as resonator reflectors has facilitated the construction of a narrow-line OPO without the aid of extra elements [15]. However, linewidth control and spectral tuning of those OPOs have been based on the use of passive elements operated in slow mechanical tuning schemes [14, 15].

In chapter 3, we demonstrate a device that integrated two PPLN EO PMCs with a PPLN optical parametric gain medium (OPGM) in a monolithic LiNbO_3 to achieve unique spectral manipulations in an OPO system. This demonstration, to the best of our

knowledge, is the first result of a pulsed narrow-line OPO whose spectral narrowing (and tuning) mechanism is electro-optically active and directly built in the same substrate along with the OPGM. The monolithic design provides lower insertion loss compared to conventional schemes using a separate linewidth-control [14]. Compared with the OPOs using the slow mechanical tuning schemes [14, 15], our device show the fast EO tuning of the narrowed spectral peak. The detail of the manufacturing process of the device can be consulted with [16].

With the EO effect (adopted in chapter 3) in annealed proton-exchanged (APE) LiNbO₃ waveguides, the coupling of a TM waveguide mode to a TE radiation mode is readily attainable due to the automatic longitudinal-phase-matching. Such a process is described in Figure 1-7. Figure 1-7 shows the effective index distributions scheme of TM and TE modes of our APE LiNbO₃ waveguide. A dot in Figure 1-7 indicates the guided mode TM₀₀, while the hatched region represents the radiation modes. For an electric field along the crystallographic y axis of our APE LiNbO₃ waveguide without any poling structure, the guided TM₀₀ mode is readily coupled to the TE mode which has propagation constant equals that of the guided TM₀₀ mode. Such a mode conversion is called an automatic longitudinal-phase-matching process. Because of the leaky nature of radiation modes, such a mode coupling process can be used to construct an active direct amplitude modulator in just single waveguide.

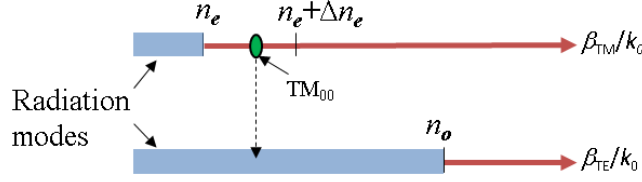


Figure 1-7. Effective index distributions scheme of TM and TE modes of our APE LiNbO₃ waveguide. n_e and n_o represent the extraordinary and ordinary refractive indices of LiNbO₃ substrate, respectively. Δn_e is the increase of the extraordinary refractive index due to the APE process. The symbols β and k_0 represent the longitudinal propagation constant and the wavenumber in vacuum, respectively.

Electro-optic control of GTR mode conversion was first studied by Marcuse in LiNbO₃ (and LiTaO₃) [17]. Unfortunately, the characterized modulation efficiency from such a simple modulator was usually very low mainly due to the inefficient field overlap between the guided mode and the phase-matched radiation mode escaping to the substrate [18]. A common approach to enhance the field-overlap and therefore the coupling efficiency is to produce a reciprocal vector via the periodic modulation of the off-diagonal dielectric tensor element of the material to compensate the phase mismatch of the two modes having efficient field overlap. Such an approach has been demonstrated via EO modulating where periodic electrodes were applied along the waveguide axial direction coincident with the crystalline x or y axis (on-axis waveguide scheme) [19]. Generally a finger-type or interdigital electrode configuration has to be used to attain the periodic sign modulation of the relevant EO coefficient of the crystal [19].

In chapter 4, we report an efficient EO TM-guided to TE-radiation mode converter in a QPM-material (PPLN) waveguide in chapter 4. In this device, a reciprocal vector is provided by the periodic modulation of the off-diagonal dielectric tensor element of the APE PPLN waveguides to compensate the phase mismatch of the two modes having efficient field overlap. So an efficient mode-field overlap in the

waveguide is obtained. Such a scheme (referred to as PPLN EO GTR PMC thereafter) is plotted in Figure 1-8.

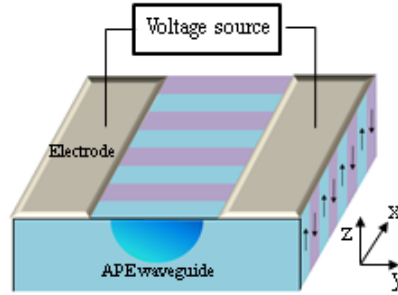


Figure 1-8. Scheme of a PPLN EO GTR PMC. x , y and z represent the principal axes. The arrows in the PPLN indicate the directions of spontaneous polarization.

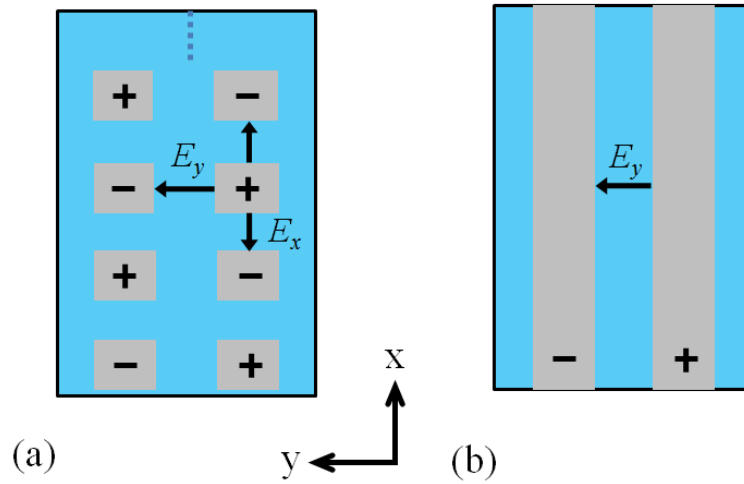


Figure 1-9. Electrode configurations for EO GTR PMCs. (a) Periodic electrodes. (b) Two continuous electrode strips.

Figure 1-9 shows two kinds of electrode configurations for EO GTR PMCs. In Figure 1-9(a), there is not only the y component of the applied electric field but also the x component of the applied electric field in periodic electrodes configuration. In contrast to the periodic electrodes configuration, our device use two continuous electrode strips configuration which supply the E_y efficiently. So not only the efficient

mode-field overlap but also the efficient E_y in the designed waveguide is obtained. The detail of the manufacturing process of the device can be consulted with [20].

Finally, chapter 5 concludes the dissertation with a conclusion of this work and suggestions for future study.

Chapter 2

Finite beam Bragg diffraction in periodically poled lithium niobate electro-optic grating

In this chapter, we report the experimental and theoretical result of the Bragg diffraction behavior of a Gaussian beam incident upon an electro-optic (EO) volume grating made of a periodically poled lithium niobate (PPLN) crystal. The theoretical result was obtained from a model based on the coupled-wave theory and a decomposition into plane waves of the incident wave.

2.1 Theoretical development

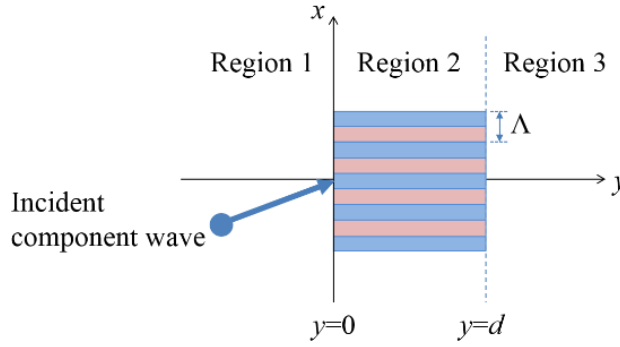


Figure 2-1. Schematic of a PPLN EO Bragg grating used as an illustration for our theoretical development.

Figure 2-1 shows the schematic of a PPLN EO Bragg grating device having a grating length of d and a period of Λ with the grating vector ($K=2\pi/\Lambda$) parallel to the crystallographic x axis of the PPLN. When driven by an external electric field E_z along the crystallographic z axis of the PPLN, a periodically modulated (relative) permittivity along the crystallographic x axis will be produced in the device and it can be expressed as

$$\varepsilon''(x) = \varepsilon_0'' + \sum_{h=1}^{\infty} \varepsilon_h'' \cos(hKx), \quad (2.1)$$

where ε_0'' is the average relative permittivity in the grating region (three regions have been defined in this theoretical treatment, as shown in Figure 2-1. The grating region corresponds to the region 2) and

$$\varepsilon_h'' = 8n_e \Delta n_e \frac{\sin(h\pi/2)}{h\pi} \quad (2.2)$$

is the Fourier amplitude of the h -th harmonic component of $\varepsilon''(x)$ for the square-wave phase grating structure, which is a function of the EO induced extraordinary refractive index change $\Delta n_e = -n_e^3 r_{33} E_z / 2$ if a z -polarized electromagnetic wave is considered (where n_e is the extraordinary refractive index and r_{33} is the relevant EO coefficient; a value of ~ 31 pm/V is used in the near-infrared).

First we assume a z -polarized plane wave of wavelength λ is incident upon such a device, the field of the wave in the grating is then governed by the wave equation as

$$\nabla^2 E''(x, y) + k^2 \varepsilon''(x) E''(x, y) = 0, \quad (2.3)$$

where $k = 2\pi/\lambda$ is the wavenumber in vacuum. To solve the equation, we expand $E''(x, y)$ in terms of the space harmonic components (with the “expansion with wavevector of magnitude equal to undiffracted wavevector” method [9]) as

$$E''(x, y) = \sum_{i=-\infty}^{\infty} A_i''(y) e^{-j(k_x'' - iK)x} e^{-j[k''^2 - (k_x'' - iK)^2]^{1/2} y}, \quad (2.4)$$

where A_i'' is the amplitude of the i -th space-harmonic component of the field

$E''(x, y)$, k_x'' is the x -component of the wave vector of the 0-th space-harmonic component in the grating, and $k'' = n_e k$ is the wavenumber of the refracted wave in the grating in the absence of permittivity modulation (i.e. at $E_z=0$). Substituting

Eq. (2.4) for $E''(x, y)$ and Eq. (2.1) for $\varepsilon''(x)$ in the wave equation (Eq. (2.3)), we

arrived at a set of one dimensional coupled-wave equations:

$$\frac{dA_i''(y)}{dy} = \frac{-j\pi^2}{g_i''\lambda^2} \sum_{h=1}^{\infty} \left[\varepsilon_h'' e^{j(g_i'' - g_{i-h}'')y} A_{i-h}''(y) + \varepsilon_h'' e^{j(g_i'' - g_{i+h}'')y} A_{i+h}''(y) \right] \quad (2.5)$$

under the slowly varying envelope approximation, where $g_i'' = [k''^2 - (k_x'' - iK)^2]^{1/2}$.

Now consider a z-polarized Gaussian beam with a waist radius of w_0 impinging on the device. The field distribution of the incident Gaussian beam at $y=0$ can be expressed in terms of a linear superposition of plane waves as

$$E'(x, y=0) = \sum_m A_m' e^{-jk_{mx}'x}, \quad (2.6)$$

where A_m' and k_{mx}' are the amplitude and the x-component of the wave vector of the m -th constituent plane wave element in region 1, respectively, and they can be obtained by taking fast Fourier transform (FFT) of the incident Gaussian beam at $y=0$. With the same coupled-wave expansion method as that used in Eq. (2.4), the corresponding electric field of the m -th constituent plane wave element of the incident Gaussian beam in the grating (the region 2) can be expressed as

$$E_m''(x, y) = \sum_i A_{m,i}''(y) e^{-j(k_{mx}'' - iK)x} e^{-j[k''^2 - (k_{mx}'' - iK)^2]^{1/2}y}, \quad (2.7)$$

where $A_{m,i}''$ is the amplitude of the i -th space-harmonic component of the m -th plane wave element and k_{mx}'' is the x-component of the wave vector of the 0-th space-harmonic component of the m -th plane wave element in the grating. The coupled-wave equations for the space-harmonic components of the m -th plane wave element of the incident Gaussian beam interacting in the grating can then be obtained by substituting Eqs. (2.7) and (2.1) in Eq. (2.3) as

$$\frac{dA_{m,i}''(y)}{dy} = \frac{-j\pi^2}{g_{m,i}''\lambda^2} \sum_{h=1}^{\infty} \left[\varepsilon_h'' e^{j(g_{m,i}'' - g_{m,i-h}'')y} A_{m,i-h}''(y) + \varepsilon_h'' e^{j(g_{m,i}'' - g_{m,i+h}'')y} A_{m,i+h}''(y) \right], \quad (2.8)$$

where $\mathcal{G}_{m,i}'' = [k''^2 - (k_{mx}'' - iK)^2]^{1/2}$. Here we define a parameter $D = \mathcal{G}_{m,1}'' - \mathcal{G}_{m,0}''$ as the dephasing factor, which is equal to zero when a plane wave element is incident at the first-order Bragg angle (i.e., at $\theta_B = \sin^{-1}(\lambda / 2n_e\Lambda)$) of the device.

Eq. (2.8) (and therefore Eq. (2.7)) can be solved by using a fourth-order Runge-Kutta algorithm with the aid of the following phase-matching and boundary conditions: First, each plane wave element field in region 1 must phase-match to the 0-th space harmonic field of the corresponding plane wave element in region 2 at $y=0$, i.e.,

$$k_{mx}'' = k_{mx}' . \quad (2.9)$$

Second, the tangential electric fields must be continuous at the boundary $y=0$, which yields

$$A_{m,i=0}''(y=0) = A_m' \text{ and } A_{m,i \neq 0}''(y=0) = 0. \quad (2.10)$$

Finally we arrived at an expression for the output electric field in region 3 as

$$E'''(x, y) = \sum_m \sum_{i=-\infty}^{i=\infty} A_{m,i}''' e^{-j[k_{mx,i}''x + k_{my,i}''(y-d)]}, \quad (2.11)$$

where $A_{m,i}'''$ is the amplitude of the i -th transmitted space-harmonic wave of the m -th plane wave element of the incident Gaussian beam and $k_{mx,i}'''$ and $k_{my,i}''' = (k''^2 - k_{mx,i}''^2)^{1/2}$ are the corresponding x - and y -components of the wave vector in this region, respectively. With the solved electric fields in region 2 (Eqs. (2.7) and (2.8)), $A_{m,i}'''$ and $k_{mx,i}'''$ can be readily derived from the relevant phase-matching and boundary conditions, i.e., each space-harmonic field of the m -th plane wave element in region 2 must phase-match to its transmitted field in region 3, which yields

$$k_{mx,i}''' = k_{mx}'' - iK, \quad (2.12)$$

and the tangential electric fields at the boundary $y=d$ must be continuous, i.e.,

$$A'''_{m,i}(y=d) = A''_{m,i}(y=d) e^{-j \left[k''^2 - (k''_{mx} - iK)^2 \right]^{1/2} d}. \quad (2.13)$$

The resultant electric field (or the diffraction field distribution) in the observation plane can then be obtained by the superposition of all the transmitted component waves reaching the plane according to Eq. (2.11).

We developed computer codes based on the above theoretical treatment to calculate and study the diffraction behavior of a 1064-nm Gaussian beam interacting with a PPLN EO Bragg grating device, in which the experimental results were compared and analyzed. In order to save the computation time and obtain a satisfactorily accurate results, we considered only the first four ($h=1-4$) harmonic terms in Eq. (2.1) and five space harmonic components ($-2 \leq i \leq 2$) for each plane wave element decomposed from the incident Gaussian beam in the built model in this work.

2.2 Other EO induced effects

To have a more comprehensive understanding of the diffraction behavior of the PPLN EO Bragg device, we discussed some effects other than those arising from the Bragg coupling mechanism studied above because they were found (see the experimental characterization section below) to play a significant role in the performance of the device especially at high working voltages. The dominant mechanism responsible for these effects can be understood by examining the basic process of the Bragg diffraction in the present device interacting with a plane wave. As illustrated in Figure 2-2, a portion of the incident wave (with an incident angle θ_{B1} corresponding to the first-order Bragg angle incident from region 1) refracts into one of the PPLN domain layers of higher refractive index (referred to as the “high-refractive-index layer” with a refractive index $n_H(V)$) at a given applied voltage of V . For simplicity, the refractive

index change at the domain boundaries of the PPLN crystal is assumed to be stepwise under the application of E_z .

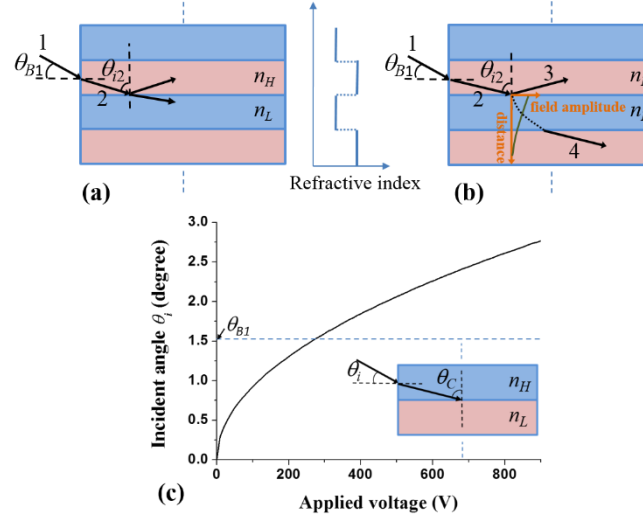


Figure 2-2. Development of a beam propagating in a high-refractive-index domain layer and passing through the dielectric interface between the layer and its neighbor low-refractive-index layer in a PPLN EO grating when the incident angle (a) $\theta_{B1} > \theta_c(V)$ or (b) $\theta_{B1} < \theta_c(V)$. (c) (Calculated) specific incident angle ($\theta(V)$) of the beam that leads to its refracted beam incident at the critical angle $\theta_c(V)$ of the dielectric interface as a function of the applied voltage.

The refracted beam (the “path 2” beam shown in Figure 2-2) then impinges on the interface between the high-refractive-index layer and a layer of lower refractive index (referred to as the “low-refractive-index layer” with a refractive index $n_L(V)$) with an incident angle of $\theta_{12}(V)$. There are two possible developments for the beam propagating in such a structure depending on the magnitude of the applied voltage, i.e., depending on whether the $\theta_{12}(V)$ is smaller or larger than the critical angle $\theta_c(V)$ ($\equiv \sin^{-1}(n_L(V)/n_H(V))$) of the dielectric interface; Figures 2-2(a) and 2-2(b) illustrate such schemes, respectively. Figure 2-2(c) shows the calculated specific incident angle $\theta(V)$ of a beam incident at the high-refractive-index layer that leads to its refracted beam

incident at the critical angle $\theta_C(V)$ of the dielectric interface as a function of the applied voltage. For the case of $\theta_{B1} > \theta_i(V)$ (leading to $\theta_{i2} < \theta_C(V)$), the refracted beam will exhibit partial reflection and transmission at the dielectric interface (see Figure 2-2(a)), which is just the typical scheme used to develop the Bragg diffraction. Here we shall focus on the particular situation when $\theta_{B1} < \theta_i(V)$ (leading to $\theta_{i2} > \theta_C(V)$), in which the refracted beam will experience the so called frustrated total internal reflection (FTIR) at the dielectric interface (refer to Figure 2-2(b)). Under this condition, the refracted beam (the “path 2” beam), instead of being totally reflected at the interface, can couple some of its energy to the next high-refractive-index domain layer (to the “path 4” beam, as illustrated in Figure 2-2(b)) via the evanescent field induced at the dielectric interface. The coupled beam can thus proceed to propagate and interact in the grating via the FTIR process occurring at the subsequent dielectric interfaces in the grating. Similar coupling process can also be developed by the reflected beam of the “path 2” beam (i.e., the “path 3” beam denoted in Figure 2-2(b)). Note no FTIR will occur along the path of the beam propagating in the present Bragg structure if the incident wave initially refracts into a low-refractive-index domain layer (no matter what the applied voltage is). Waves involving with such a FTIR coupling mechanism can contribute to the Bragg diffraction process and can still be possible to be observed in the present device if the evanescent coupling strength is strong enough (see below). Since the evanescent field is damped exponentially along the direction normal to the dielectric interface, the strength of the wave coupling in the process shown in Figure 2-2(b) will depend on a penetrating distance q of the evanescent field. This distance q is defined by the distance where the field amplitude is attenuated to $1/e$ of its maximum amplitude and given by [21]

$$q(V) = \frac{\lambda / (2\pi n_L(V))}{\{(n_H(V) / n_L(V))^2 \sin^2(\theta_{i2}(V)) - 1\}^{1/2}}. \quad (2.14)$$

Figure 2-3 shows the calculated penetrating distance q as a function of the applied voltage according to Eq. (2.14).

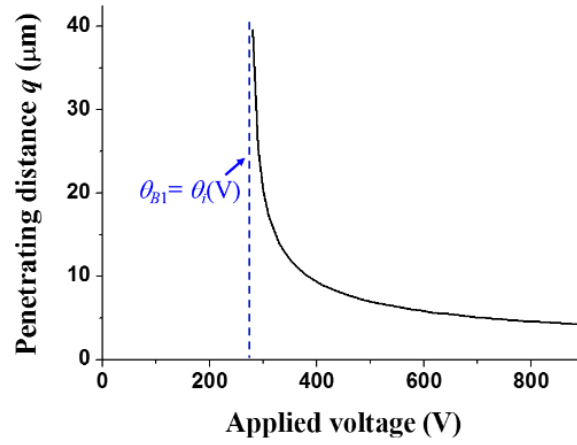


Figure 2-3. Calculated penetrating distance q as a function of the applied voltage. The dashed blue line denotes the voltage in which the device operates at the situation with $\theta_{B1} = \theta_i(V)$.

Since the thickness of each domain layer in our PPLN EO Bragg grating is 10 μm , it is interesting to examine the situation when the q value of the evanescent field is smaller than 10 μm . It can be expected in this situation the evanescent coupling strength will be relatively weak, implying the refracted beam (the “path 2” beam in Figure 2-2(b)) will tend to be confined in the high-refractive-index domain layer and be guided along the crystallographic y direction. Such a phenomenon is called light-guiding effect [22]. As a reference, a voltage of $>390\text{V}$ is required to lead to the result of $q < 10 \mu\text{m}$ according to Figure 2-3 for this illustrative scheme (i.e., a scheme of a plane wave interacting with a square-wave index profile grating). This effect is believed to be the major origin of a background light distribution found besides the Bragg diffraction mode distribution in our device especially at high applied voltages (see below).

2.3 Device and experimental setup

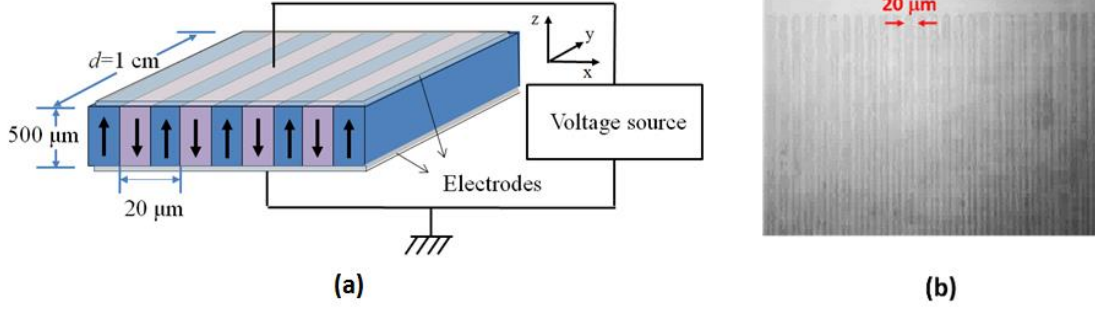


Figure 2-4. (a) Schematic configuration of a PPLN EO grating device. (b) A typical microscopic image of the HF-etched +z surface of the fabricated PPLN crystal.

We fabricated the PPLN EO Bragg device in a 1-cm long, 500- μm -thick z-cut LiNbO_3 crystal using the standard electric-field poling technique [23] (refer to Figure 2-4(a) for the schematic of the device). The poled grating period Λ is $20\text{ }\mu\text{m}$, leading to a Bragg angle of $\theta_B \sim 0.7^\circ$ (in the crystal). The choice of $20\text{ }\mu\text{m}$ as the grating period for this work was because it is large enough for the fabrication of a PPLN grating with relatively good domain wall quality using today's electric-field poling technique. On the other hand, an even larger grating period might not be preferred because a thicker domain length will facilitate the occurrence of the unwanted light-guiding effect at a relative small operating voltage as discussed in the preceding section. Figure 2-4(b) shows a typical microscopic image of the HF-etched +z surface of the fabricated PPLN. The domain walls look uniform and straight in this observation scale. The domain duty cycle approaches 50%. To reduce the stress-induced refractive index change at the PPLN domain boundaries, this device has been annealed at a temperature of 500°C for 5 hours [4]. The two y faces of the crystal were optically polished without the application of a coating layer. The $\pm z$ surfaces of the crystals were sputtered with NiCr alloy as electrodes for the E_z application. The diffraction behavior of the fabricated PPLN EO Bragg grating was then experimentally characterized with a Gaussian-beam

laser to compare with that predicted by the theoretical model developed in this work. The laser was a continuous-wave TEM₀₀ mode ($M^2 \sim 1.1$) ytterbium fiber laser emitting at 1064 nm. As shown in Figure 2-5, the laser beam was first collimated and then passed through a polarizing beam splitter followed by a lens system to produce an extraordinarily polarized wave of proper beam size before entering the PPLN Bragg device. The beam was incident at the first Bragg angle of the device, which is about 0.7° inside the PPLN grating at 26°C. The laser beam waist was located at near the front surface of the crystal (at $y=0$; see Figure 2-1). In this experiment, we have investigated the diffraction behavior of the device for two different beam sizes (which were 28 and 135 μm in waist radius ($1/e^2$ half-width), respectively, deduced by the 10/90 knife-edge method). The PPLN EO Bragg grating was placed in a crystal holder where both the temperature control and voltage application of the device can be performed. The holder was mounted on a 5-axis micro-positioning stage. A beam profiler, placed 3.7 cm downstream from the crystal, was used to monitor the full diffraction pattern and the output power of beam after traversing the Bragg device.

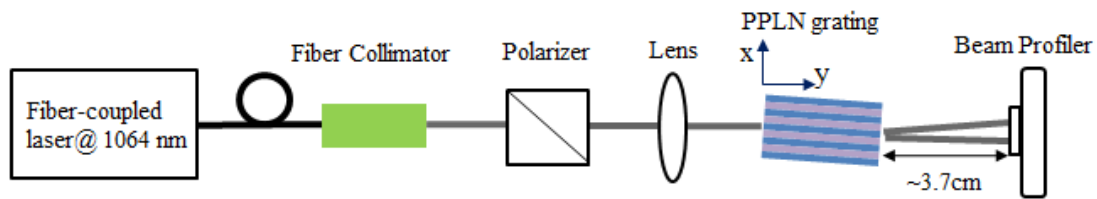


Figure 2-5. Schematic arrangement of the experimental setup.

2.4 Experimental results and theoretical simulation analysis

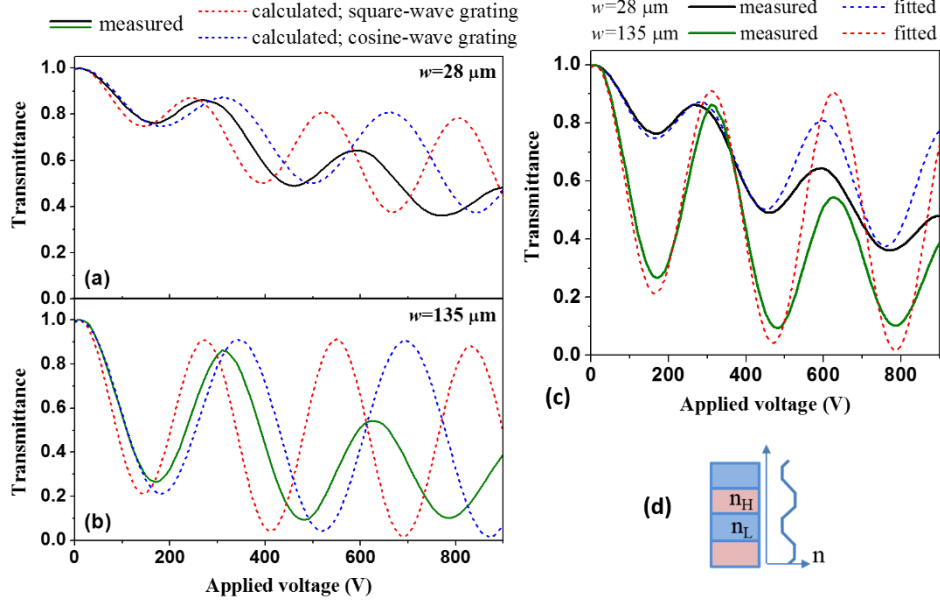


Figure 2-6. Measured (black and green curves) and calculated (dashed red and blue curves for square-wave and cosinusoidal-wave shape gratings, respectively) 0^{th} order transmittances of a 1064-nm laser after traversing the PPLN EO Bragg device as a function of the applied voltage for beam waist radii of (a) 28 and (b) 135 μm . (c) the same measurement results (black and green curves) as plotted in (a) and (b) but with simulations (dashed curves) based on a modified Fourier amplitude of the modulated permittivity (see text). (d) A periodic trapezoidal grating form.

The black and green curves in Figures 2-6(a) and 2-6(b) show the measured 0^{th} order transmittances of the 1064-nm laser after traversing the PPLN EO Bragg device as a function of the applied voltage for two different beam waist radii, 28 and 135 μm , respectively. The peak transmittance locates slightly away from 0 V due to the effect of the internal field induced refractive index change at the PPLN domain boundaries [24]. The dashed red and blue curves shown in Figures 2-6(a) and 2-6(b) represent the calculation results based on PPLN whose grating structure is assumed to be formed by a square-wave and a cosinusoidal-wave shape gratings, respectively. Here a cosinusoidal-wave shape grating is included in the calculation because it has been

shown the refractive index across the boundary of opposite polarity domains in LiNbO₃ under the EO effect could be in a continuous change instead of a stepwise change (as seen from the PPLN domain shape) [25]. We have shifted the calculation results (and all of the following calculation results made in this work) 10 V to take into account of the internal-field effect mentioned above before comparing them with the experimental results. As shown in Figures 2-6(a) and 2-6(b), the presence of considerable discrepancy between the calculated and measured results suggests that the effective refractive index profile of our fabricated PPLN EO grating is neither an ideal square-wave nor a typical cosinusoidal-wave forms. We found that we can obtain simulation curves more close to the measured transmission curves by modifying the Fourier amplitude ε_h'' of the modulated relative permittivity given in Eqs. (2-1) and (2-2) to $\varepsilon_1'' = 2.22n_e\Delta n$ and $\varepsilon_{h\neq 1}'' = 0$. The dashed curves in Figure 2-6(c) represent the simulation results with such a modification. The justification for using a larger ε_1'' in this simulation is in contrast to that with the cosinusoidal-wave grating function ($\varepsilon_1'' = 2n_e\Delta n$ and $\varepsilon_{h\neq 1}'' = 0$) has been based on two facts. First, our experimental results have shown that the Bragg conversion is only observable between the two lowest (i.e., the 0th and 1st) order beams in the present device, implying it is the 1st spatial harmonic component of the Bragg grating structure dominates the Bragg conversion process according to Eq. (2-8). Second, it can be concluded from the calculated transmission curves shown in Figures 2-6(a) and 2-6(b) that the greater the weighting of the 1st spatial harmonic component in the Bragg grating structure (as is the case with the square-wave grating shape), the lower the specific voltages that are required to reach the extreme values of the curves. And it can be seen that the specific voltages for reaching the extreme values of the experimental transmission curves are lower than

those for the curves calculated with the cosinusoidal-wave function. We found this modified amplitude of the 1st harmonic component (i.e., $\varepsilon_1'' = 2.22n_e \Delta n$) is very close to that expanded from a periodic trapezoidal waveform. This waveform (see Figure 2-6(d)) indeed exhibits a profile with a continuous refractive index change across the domain boundary [25].

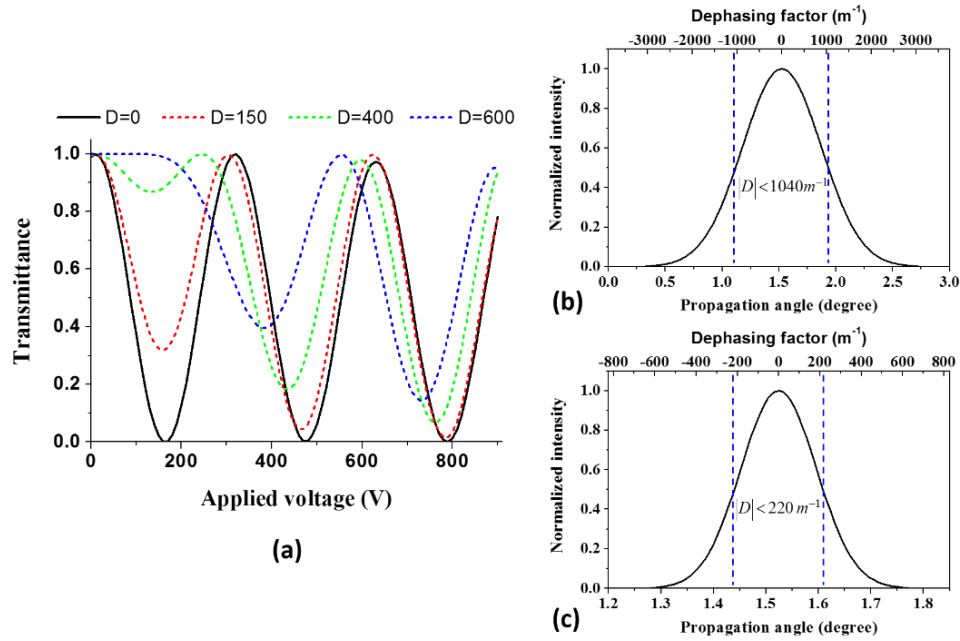


Figure 2-7. (a) transmittance of the 0th order harmonic wave as a function of the driving voltage of the PPLN EO Bragg device for plane wave elements of different absolute values of the dephasing factor D. The right Figures show the angular spectra and the corresponding dephasing factor distribution of the incident Gaussian beams with waist radii of (b) 28 and (c) 135 μm .

To understand the aperiodically damped oscillatory behavior of the transmission curves in Figure 2-6, we calculated the transmittance of the 0th order harmonic wave (defined as $|A_{m,i=0}''(y = 1 \text{ cm}) / A_{m,i=0}''(y = 0 \text{ cm})|^2$ according to Eq. (2-8)) as a function of the driving voltage of the present device for plane wave elements of different absolute

values of the dephasing factor D (see section 2.2), as shown in Figure 2-7(a). The calculations have been based on the modified index modulation function used in obtaining Figure 2-6(c). The voltage-dependent transmission behavior of the present device is then determined by a superposition effect of the transmission characteristics of all the constituent plane wave elements (of different absolute D) interacting in the device. In Figure 2-7(a), the transmission behavior with $D=0$ is similar to that predicted by the two-wave first-order coupled-wave theory [8] which has been widely used in solving a simplified Bragg diffraction problem (dealing with an infinite plane wave incident at the Bragg angle of a Bragg device) [3-7]. It is not surprising to find from Figure 2-7(a) that efficient energy conversion occurs only for those plane wave elements with a small absolute value of D (e.g., >60% conversion efficiency can be obtained for plane wave elements with $|D| < 150 \text{ m}^{-1}$ referring to the first minimum of the transmission curves occurring around 170 V) because of the finite Bragg phase-matching bandwidth. However, with the increase of the driving voltage, it can also be found from Figure 2-7(a) even plane wave elements with large absolute values of the dephasing factor can convert an appreciable amount of energy to their 1st harmonic waves (refer to those local minimums of the transmission curves occurring around 470 and 780 V) due to the enhancement of the grating strength (and therefore the wave coupling strength). And this enhancement leads to a tendency shown in Figure 2-6(c) that the higher the driving voltage, the lower the minimum transmittance that can be obtained in this device except the third minimum transmittance observed with the Gaussian beam of 135 μm beam radius around 790 V. The deviation of the experimental observation from the prediction around 790 V can be mainly attributed to the light-guiding effect (which is prominent at such high voltages) as discussed in section 2.2.

Besides the extreme values of the transmission curves, it is also interesting to find that the oscillation periods of the curves shown in Figure 2-7(a) are dependent on the absolute magnitude of the dephasing factor D of the plane wave elements as well (the larger the $|D|$, the smaller the period). Figures 2-7(b) and 2-7(c) show the angular spectra of the incident Gaussian beams with waist radii of 28 and 135 μm , respectively. The lower and upper horizontal axes represent the propagation angles θ_m ($\equiv \sin^{-1}(k'_{mx} / |k'_m|)$) in air and the corresponding dephasing factor values of the decomposed plane wave elements, respectively. As expected, a more tightly focused Gaussian beam (here with a waist radius of 28 μm) is characterized by having a wider distribution of the element dephasing factor (with a FWHM of $|D| < 1040 \text{ m}^{-1}$, see Figure 2-7(b)) in contrast to a less focused one (here with a waist radius of 135 μm and a FWHM of $|D| < 220 \text{ m}^{-1}$, see Figure 2-7(c)). Since it is the plane wave elements with $|D| < 150 \text{ m}^{-1}$ that can convert appreciable energy to their 1st harmonic waves first at $\sim 170 \text{ V}$ according to Figure 2-7(a) (see also the analysis below) and since $|D| < 150 \text{ m}^{-1}$ is well within the dephasing bandwidths of the two Gaussian beams according to Figures 2-7(b) and 2-7(c), the first minimum transmittance of the device for the two beams occurs at almost the same voltage of 170 V as shown in Figure 2-6(c). However, Figure 2-6(c) also shows that smaller voltages are required to reach the following extreme values of the transmission curve for the beam with a smaller beam radius (i.e., 28 μm). This can be understood by the fact that these extreme values occur at voltages $> 170 \text{ V}$, implying more dephasing plane wave elements (say, with $|D| > 220 \text{ m}^{-1}$) can now participate in and contribute to the Bragg conversion process as discussed earlier. Since the beam with a smaller beam radius is composed of more

dephasing elements (see Figures 2-7(b) and 2-7(c)) that are characterized by having a transmission curve with a smaller oscillation period as revealed in Figure 2-7(a), the existence of certain difference of the driving voltages required to reach the transmission extreme values between the two beams of different beam sizes in figure 2-6(c) can then be understood.

Figure 2-8 shows the measured (black curves) beam intensity distribution along the central horizontal line of the beam profiler (corresponding to the dashed lines in the insets) for various specific voltages (refer to Figure 2-6(c)) of reaching the first maximum (Figs. 2-8(a) and 2-8(b)), the first minimum (Figs. 2-8(c) and 2-8(d)), the second maximum (Figs. 2-8(e) and 2-8(f)), the second minimum (Figs. 2-8(g) and 2-8(h)), and the third minimum (Figs. 2-8(i) and 2-8(j)) transmittances of the device for the incident Gaussian beam with waist radii of 28 and 135 μm , respectively. The insets show the captured images of the corresponding 2D intensity distributions. It is interesting to find that the beam modes have deformed after traversing the crystal and become horizontal and vertical ellipse modes for the 28 and 135- μm waist-radius beams, respectively, as observed in Figures 2-8(a) and 2-8(b). This mode deformation phenomenon could be a result of the diffraction in an anisotropic media [26] and should have minor influence on our experimental investigation because only the extraordinary component waves (propagating in or near the x - y plane) which are the dominant component in our incident Gaussian beam contribute to the Bragg conversion process in our device due to the grating strength of our device.

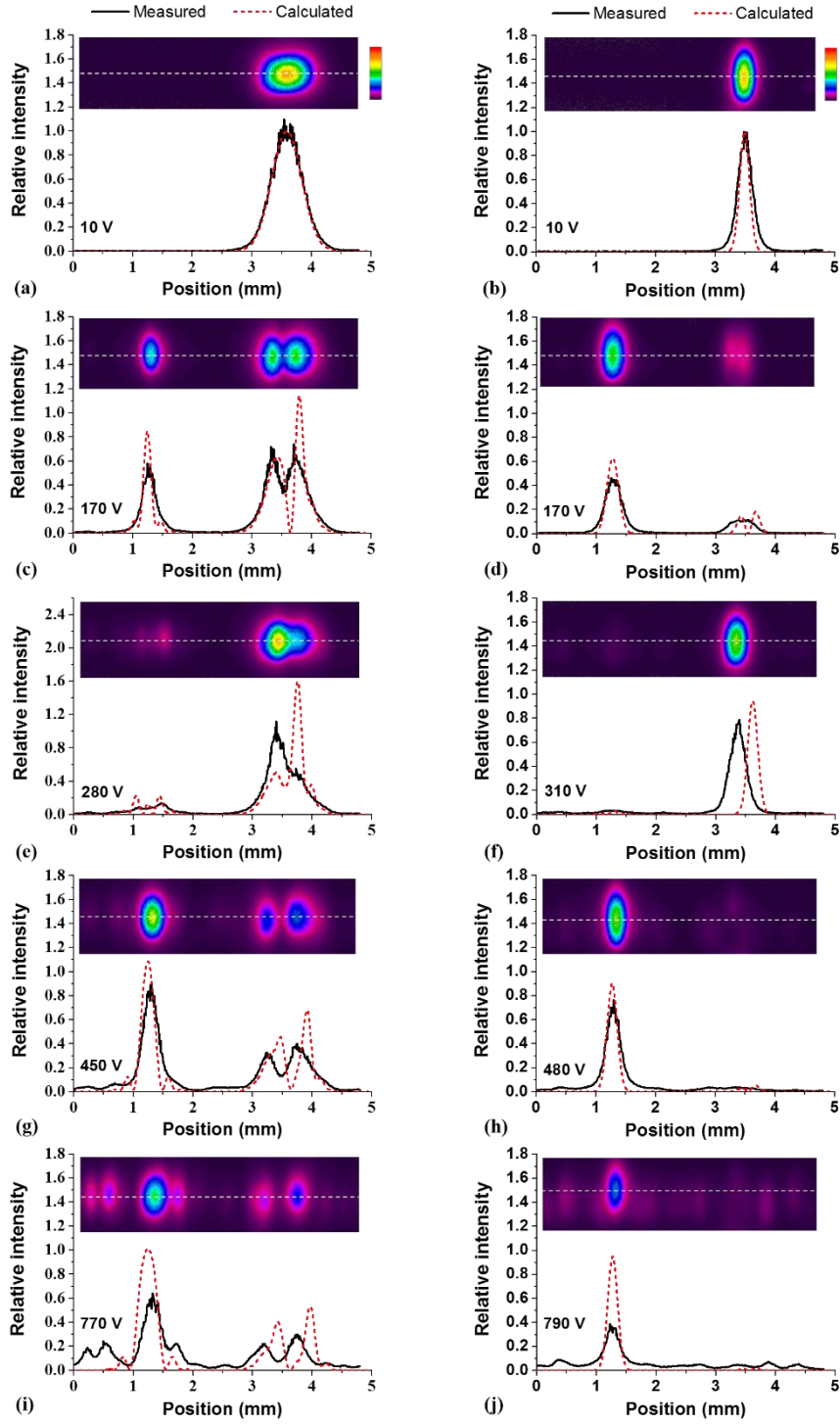


Figure 2-8. Measured (black curves) and calculated (red curves) beam intensity distribution along the central horizontal line of the beam profiler (corresponding to the dashed lines in the insets) for various specific operating voltages of the device for incident Gaussian beam waist radii of $28\ \mu\text{m}$ (left Figures) and $135\ \mu\text{m}$ (right Figures). The insets show the captured images of the corresponding 2D intensity distributions.

The red curves shown in Figure 2-8 represent the calculation results using the theoretical model developed in this work. The positions and peak intensities of the calculated diffraction patterns shown in Figures 2-8(c)-2-8(j) have been adjusted with reference to those used in fitting the measurement results of the 0th order beam at 10 V (see Figures 2-8(a) and 2-8(b)). Besides, the positions of the calculated 1th order beams were adjusted with reference to the location of the measured 1th order beams at 170 V (see Figures 2-8(c) and 2-8(d)) by tuning the simulation parameter of measurement distance. The information shown in Figure 2-8 is thus comparable (for those with the same incident beam size). Note the measured intensity distribution of the 0th order beam shown in Figure 2-8(b) has been slightly broader than the calculated one due to the existing spatial resolution limit of our beam profiler (which is $\sim 500\ \mu\text{m}$). As shown in Figure 2-8, the calculation results are in substantial agreement with the experimental results, and these results do reveal that the diffraction mode of a Gaussian beam interacting with the PPLN EO Bragg device is a function of the beam size and the applied voltage. To help analyze the results shown in Figure 2-8, we further investigated the corresponding diffraction behavior of the plane wave elements (decomposed from the incident Gaussian beams) after traversing the present device in the angular spectral domain. Figure 2-9 shows the calculated output angular spectra (with the same plot configuration as Figures 2-7(b) and 2-7(c)) for again the two Gaussian beams and those applied voltages used in Figure 2-8.

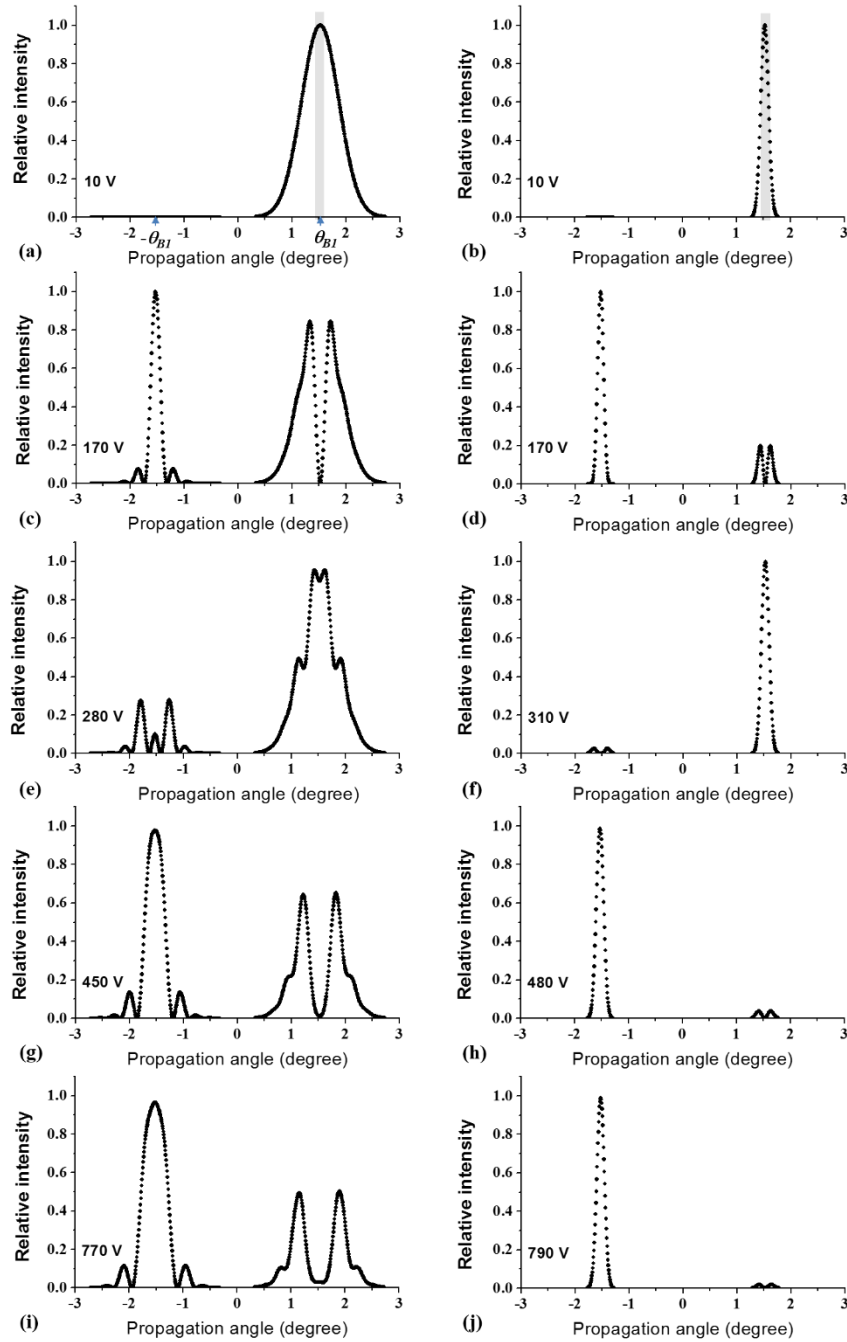


Figure 2-9. Calculated output angular spectra of two Gaussian beams of waist radii of 28 μm (left Figures) and 135 μm (right Figures) after traversing the PPLN EO Bragg device under those specific operating voltages used in figure 2-8. The vertical scales are directly comparable for those results obtained from the same incident beam size.

The angular spectra shown in Figures 2-9(a) and 2-9(b) are very similar to those obtained in Figures 2-7(b) and 2-7(c) (the input angular spectra), respectively, because

in principle no Bragg conversion could occur at this initial voltage (10 V). When operating at the first minimum transmittance of the device (at an applied voltage of ~ 170 V according to Figure 2-6(c)), it can be found from Figure 2-9(c) that only plane wave elements within the dephasing range of $|D| < 150 \text{ m}^{-1}$ (the shadowed regions shown in Figure 2-9(a)) can efficiently convert their energy to their 1st harmonic waves (centered at $\theta_m = -\theta_{B1}$) as revealed above. Similar mechanism can be found in Figure 2-9(d) for an incident Gaussian beam with a $135 \text{ }\mu\text{m}$ beam waist radius, where, however, most of the incident plane wave elements are well within the dephasing range of $|D| < 150 \text{ m}^{-1}$ and can be converted to the 1st harmonic waves.

With the aid of these angular spectra information, the observed diffraction intensity profiles shown in Figures 2-8(c) and 2-8(d) can now be easily understood. However, we find that there exists some shift of the dip/peak positions of the 0th order beam between the measured and calculated results. We suspect the shift of the beam positions could originate from the effect that some of the plane wave elements involving in the FTIR process (occurring when $\theta_2 > \theta_C(V)$ as discussed in section 2.2) have contributed to the Bragg conversion process with extra phase shifts. Further study is required to confirm the mechanism for such a beam shift effect.

Figures 2-9(e) and 2-9(f) show the angular spectra for the situations when the device operates at the second maximum transmittance for the beam with waist radii of 28 and $135 \text{ }\mu\text{m}$, respectively. Both spectra indicate that the 1st order harmonic waves within the dephasing range of $|D| < 150 \text{ m}^{-1}$ are now converted back to their 0th harmonic waves with relatively high efficiency due to again the effect of the finite Bragg conversion bandwidth. Furthermore, it is interesting to see in the spectra that some relatively weak peaks emerge at around $\theta_m = -\theta_{B1}$. These 1st harmonic waves are

converted from some plane wave elements of the 0th order beams with $|D| > 150 \text{ m}^{-1}$. It is the main reason that we observed a transmittance of less than unity at the second maxima of the transmission curves shown in Figure 2-6(c). These relatively high dephasing plane wave elements perform the 0th to the 1st order Bragg conversion (though with much less efficiency) at this operating voltage ($\sim 280 \text{ V}/\sim 310 \text{ V}$ according to Figure 2-6(c)) because they are characterized by larger phase mismatches in the Bragg conversion process and therefore have different oscillation periods of the transmission curves from those of $|D| < 150 \text{ m}^{-1}$ as shown in Figure 2-7(a). Again the measured diffraction intensity profiles shown in Figures 2-8(e) and 2-8(f) can be understood with what has been analyzed above with the calculated angular spectra except the existence of the shift of the peak positions of the 0th order beam between the measured and calculated results, which has been observed in Figures 2-8(c) and 2-8(d).

In contrast to the angular spectra shown in Figures 2-9(c) and 2-9(d), the angular spectra shown in Figures 2-9(g) and 2-9(h) derived from the second minima in the transmission curves in Figure 2-6(c) (at driving voltages of ~ 450 and $\sim 480 \text{ V}$, respectively) show that plane wave elements of a broader range of the dephasing factor can convert appreciable energy to their 1st harmonic waves due to the increased grating strength. The enhancement of coupling strength of the dephasing elements increases the intensity of the 1st order beams observed in Figures 2-8(g) and 2-8(h) in contrast to those shown in Figures 2-8(c) and 2-8(d). The calculation results of the angular spectra shown in Figures 2-9(i) and 2-9(j) suggest an even higher conversion efficiency due to the aforementioned grating strength enhancement effect when the device operates at the third minima in the transmission curves in Figure 2-6(c) (at voltages of ~ 770 and $\sim 790 \text{ V}$, respectively). However, we found the measured peak powers of the 1st order beams shown in Figures 2-8(i) and 2-8(j) are instead smaller than those corresponding

ones shown in Figures 2-8(g) and 2-8(h), respectively. Besides, the background intensity levels in Figures 2-8(i) and 2-8(j) are higher than those measured at the second minimum transmittances of the device (see Figures 2-8(g) and 2-8(h)). These observation results can be attributed to the presence of strong light-guiding effect at such high operating voltages as discussed in section 2.2. The light-guiding effect will lead to lower Bragg diffraction efficiency because the effect will confine the beam power in those high-index layers and form a background light distribution without following the Bragg law.

2.5 Chapter summary

The finite beam Bragg diffraction characteristics of a PPLN EO grating device were for the first time to the best of our knowledge investigated both theoretically and experimentally. A theoretical model has been developed based on the coupled-wave theory with the aid of the PWD method to obtain the coupled-wave equations as well as to help analyze the experimentally diffraction behavior of the PPLN EO Bragg device. From the analysis, we found the grating strength and the dephasing degree of the plane wave elements that determine the finite-beam diffraction characteristics of the PPLN EO Bragg device. As increasing the applied voltage, there are more and more dephasing plane wave elements that have ability to convert its appreciable energy between the 1st and 0th order harmonic waves. Due to the different phase-mismatching conditions of these dephasing plane wave elements, they convert their appreciable energy between the 1st and 0th order harmonic waves with different oscillating periods. Besides, some unique EO induced effects found in the PPLN grating such as the light guiding effect and the FTIR wave coupling process were suggested to be responsible

for the formation of a background light distribution in the diffraction pattern and the slight deviation of the diffraction mode position from that predicted, respectively.

Chapter 3

An optical parametric oscillator using periodically poled lithium niobate electro-optic polarization-mode converters

Optical parametric oscillators (OPOs) are popular sources for producing wavelength-tunable coherent radiation for various applications. Besides the tunability, many applications such as spectroscopy, communication, LIDAR, and remote sensing demand light sources working at a narrow linewidth. Thus, a spectral narrowing mechanism needs to be introduced to the OPO (especially for pulsed OPOs characterized by a broader bandwidth). In this chapter, we have integrated two PPLN EO PMCs with a PPLN optical parametric gain medium (OPGM) in a monolithic LiNbO_3 to meet the requirements of these applications in an OPO system.

3.1 Device

We fabricated the integrated PPLN crystal in a 55 mm long, 1 mm wide, and 0.5 mm thick z-cut LiNbO_3 crystal by using the standard electric-field poling technique [23]. Figure 3-1 shows the schematic arrangement of the integrated PPLN device comprising a 3 cm long PPLN OPGM and two 1.25 cm long PPLN EO PMCs (referred to in the Figure labels as the first and second PPLN EO PMCs, respectively). The PPLN EO PMCs have a grating period of $21\text{ }\mu\text{m}$ phase-matched to the codirectionally coupled $\tilde{\text{Solc}}$ -type Bragg condition at 1532 nm at 40 °C [13]. The PPLN OPGM is sandwiched between the two PPLN EO PMCs and has a grating period of $29.8\text{ }\mu\text{m}$ phase-matched to the 1064 nm pumped 1532 nm signal and the 3483 nm idler parametric generation process at 40 °C [27]. Both of the end (x) faces of this crystal received an optical-grade

polish and an antireflection (AR) coating for both 1064 nm and 1532 nm. The two y faces of the PPLN EO PMCs were sputtered with NiCr alloy to form the side electrodes.

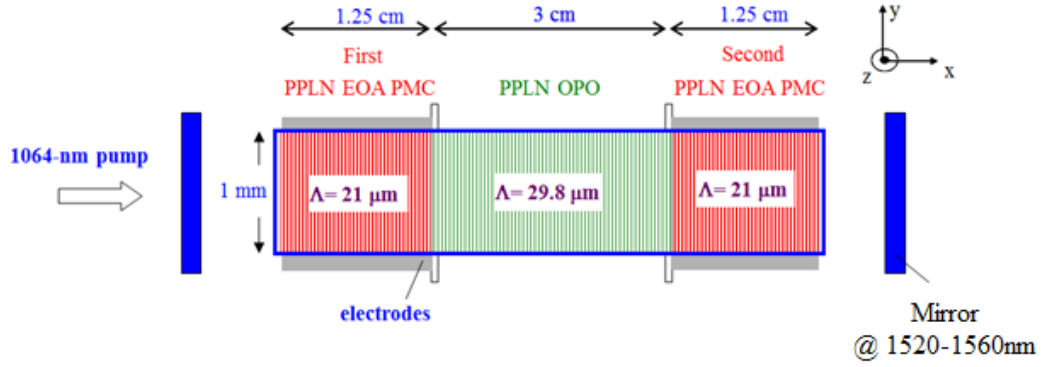


Figure 3-1. Schematic arrangement of the 1064 nm pumped OPO system at the 1532 nm band based on a monolithic LiNbO₃ integrating a 3 cm long, 29.8 μm period PPLN OPGM and two 1.25 cm long, 21 μm period PPLN EO PMCs.

3.2 Theoretical analysis

In this device, the PPLN OPGM generates and amplifies only the extraordinary (e) polarized waves satisfying the eee QPM condition. Under the slowly envelop varying approximation, a three-plane-wave parametric interaction of the PPLN OPGM involving angular frequencies ω_p , ω_s and ω_i are described by the coupled wave equations [1]

$$\begin{aligned} \frac{dE_{s,e}}{dx} &= -i \frac{\omega_s d_Q}{n_{s,e} c} E_{p,e} E_{i,e}^* \exp(-i\Delta k_Q x) \\ \frac{dE_{i,e}}{dx} &= -i \frac{\omega_i d_Q}{n_{i,e} c} E_{p,e} E_{s,e}^* \exp(-i\Delta k_Q x), \\ \frac{dE_{p,e}}{dx} &= -i \frac{\omega_p d_Q}{n_{p,e} c} E_{s,e} E_{i,e} \exp(i\Delta k_Q x) \end{aligned} \quad (3.1)$$

The symbols n , c , k , and E represent the refraction index, light speed in vacuum, wavenumber, and amplitude of the extraordinary polarized wave, respectively.

The subscripts p, s, i, and e in the equations stand for the pump, signal, idler, and extraordinary wave, respectively. d_Q is the effective nonlinear coefficient for the QPM interaction and is expressed by

$$d_Q = d_{33} \frac{2}{\pi}, \quad (3.2)$$

where d_{33} is the involving nonlinear coefficient, $2/\pi$ is the first order Fourier coefficient for a 50% duty cycle in the QPM structure. Δk_Q is the wave vector mismatch for the QPM interaction of the PPLN OPGM and is expressed by

$$\Delta k_Q = k_{p,e} - k_{s,e} - k_{i,e} - \frac{2\pi}{\Lambda_{\text{OPGM}}}, \quad (3.3)$$

where Λ_{OPGM} is the period of the PPLN OPGM.

In contrast to the PPLN OPGM, the PPLN EO PMCs of our device perform narrowband e $\rightarrow$ ordinary (o) polarization-mode conversion at the signal wave and can be described by the coupled wave equations [12]

$$\begin{aligned} \frac{dE_{s,e}}{dx} &= -i\kappa^* E_{s,o} e^{-i\Delta kx} \\ \frac{dE_{s,o}}{dx} &= -i\kappa E_{s,e} e^{i\Delta kx}, \end{aligned} \quad (3.4)$$

where $E_{s,e}$ and $E_{s,o}$ are the amplitudes of the ordinary mode and the extraordinary mode of the signal wave, respectively. Δk is the wave vector mismatch for the QPM interaction of the PPLN EO PMC and is expressed by

$$\Delta k = k_{s,o} - k_{s,e} - \frac{2\pi}{\Lambda_{\text{PMC}}}, \quad (3.5)$$

where Λ_{PMC} is the period of the PPLN EO PMC. The coupling coefficient κ is defined as

$$\kappa = \frac{\omega_s}{c} \frac{n_{s,o}^2 n_{s,e}^2 r_{51} E_y}{\pi \sqrt{n_{s,o} n_{s,e}}}, \quad (3.6)$$

where r_{51} is the electro-optic coefficient (a value of ~ 26 pm/V is used in the near-infrared [28]) and E_y is the applied field along the crystallographic y axis. Since

the conversion spectrum of a PPLN EO PMC depends on the driving field and the temperature (see below), it is possible to produce a specific gain spectrum for the signal oscillation depending on the overlap of the conversion spectra of the EO PMCs and the OPGM in the presented scheme. The conversion bandwidth (FWHM) of a PPLN EO PMC can be derived from [12], given by

$$\Delta\lambda_{\text{EOPMC}} = \frac{2\lambda_C}{\pi} \frac{\sqrt{\left[\text{sinc}^{-1}(1/\sqrt{2}\kappa L_C)\right]^2 - (\kappa L_C)^2}}{t_C L_C} \times \left| \frac{n_{s,o} - n_{s,e}}{\lambda_C} - \frac{\partial(n_{s,o} - n_{s,e})}{\partial\lambda_C} \right|^{-1}, \quad (3.7)$$

where λ_C is the working wavelength, L_C is the crystal length, t_C is the number of transits through the crystal. Since κ is a function of the external electric field E_y , EO control of the conversion spectrum of a PPLN EO PMC is feasible according to Eq. (3.7).

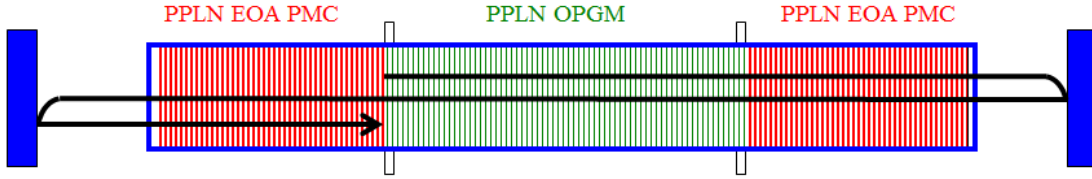


Figure 3-2. The first round trip of the buildup process of the OPO signal.

Considering the buildup process of the OPO signal in the first round trip shown in Figure 3-2, the signal wave generated from the PPLN OPGM will interact with the second PPLN EO PMC first, followed by another interaction with the PMC after the mirror reflection (the practice of a PPLN EO PMC in a resonator has been reported elsewhere [28]). The backward-propagating signal wave then traverses the PPLN OPGM without experiencing any gain because of the absence of a pump field for the backward direction. The propagation through the device continues with two successive interactions with the first PPLN EO PMC via a mirror reflection. The signal wave then again enters the PPLN OPGM in a forward direction, where only the e-polarized

spectral modes will be amplified. Here, the first round trip in the resonator ends. Figure 3-3 shows the calculated e-polarized wave spectra after the first round trip in the device at the lowest and a higher EO driving field (same fields for both PMCs) for 100% conversion at 1532 nm.

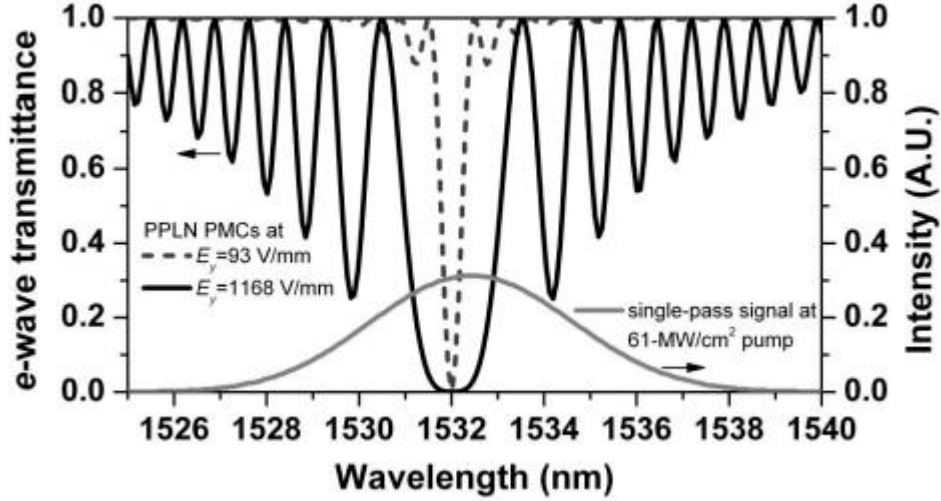


Figure 3-3. Calculated e-polarized wave spectra after the first round trip in the device shown in Figure 3-2 at $E_y = 93$ V/mm(dotted line) and 1168 V/mm (solid black line), provided a flat-spectrum signal wave is generated from the OPGM. Both PMCs are operated with identical driving fields. The solid gray line represents the actual single-pass signal spectrum with the PPLN OPGM pumped by a 61 MW/cm² 1064 nm wave.

In the case of a higher EO driving field, it provided a flat-spectrum signal wave. The actual single-pass signal spectrum with the PPLN OPGM pumped by a 61 MW/cm² 1064 nm wave is displayed as the gray (lower) line. As revealed in Figure 3-3, the higher the E_y is, the broader the central bandwidth and the larger the sidelobe ringing level and period of the notch-filter-like spectrum formed by the PPLN EO PMCs. At an appropriate E_y , such a notch-filter-like spectrum works to effectively narrow and tailor the phase-matching spectrum of the OPGM for single or multiple signal spectral peak generation. Specifically, the two PMCs can work at different

driving fields to allow the composition of diverse filtering spectra. Besides, temperature control of the system can be utilized to facilitate the tailoring of a desired signal gain spectrum via the shifts of the phase-matching spectra in a tuning rate of $+0.24 \text{ nm/}^{\circ}\text{C}$ for the PPLN OPGM and $-0.785 \text{ nm/}^{\circ}\text{C}$ for the PPLN EO PMC [13].

The OPO spectra results from the overlapping line shape functions of the OPGM and the intracavity linewidth-control element(s) after multiple transits in the resonator [14]. Those spectra cannot be predicted phenomenologically. We therefore developed an algorithm to simulate the signal field evolution in the OPO by numerically solving the respective coupled-wave equations (3.1) and (3.4) for each PPLN section. This is done repeatedly to account for the resonant signal buildup during a number of cavity transits.

3.3 Experimental setup

Figure 3-4 is a schematic drawing of the experimental setup. The fabricated PPLN crystal was mounted in a three-zone temperature-controlled oven and positioned in a linear resonator formed by two identical 100 mm radius-of-curvature BK7 glass (a borosilicate crown glass) mirrors having coating layers of $\sim 95\%$ reflectance at 1520 – 1560 nm and $>99.9\%$ transmittance at 1064 nm. The operating temperatures of the two EO PMCs can be separately controlled and detuned by 3°C from that of the PPLN OPGM. The pump source is a linearly polarized, actively Q-switched Nd:YAG laser generating 10 Hz repetition rate 1064 nm pulses of duration $\sim 7 \text{ ns}$.

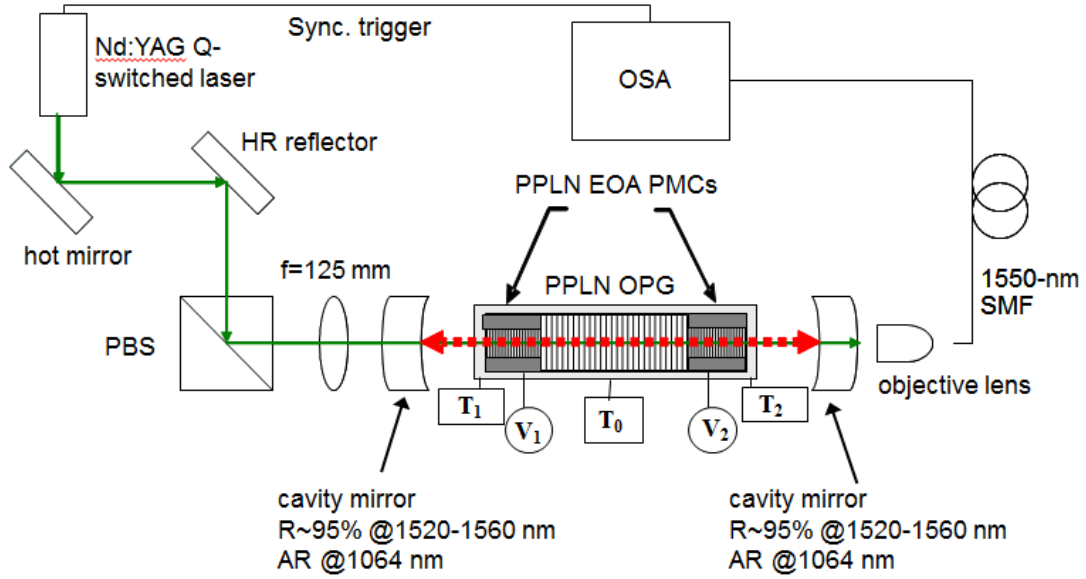


Figure 3-4. Schematic drawing of the experimental setup.

The peak pump power input to the OPO system is ~ 23 kW, exceeding the observed threshold power (~ 4 kW) by ~ 6 times. This threshold agrees with theoretical estimations (~ 3.3 kW) [14], assuming system losses originate only from the mirror couplings. The pump beam size in the crystal is ~ 220 μm . An optical spectrum analyzer with a spectral resolution of 0.01 nm was used to measure the signal output spectra.

3.4 Experimental results

Figures 3-5 and 3-6 show the measured (solid black line; blue/red on line) and calculated (dashed gray line) output signal spectra from the OPO system using the unique spectral manipulation technique introduced above.

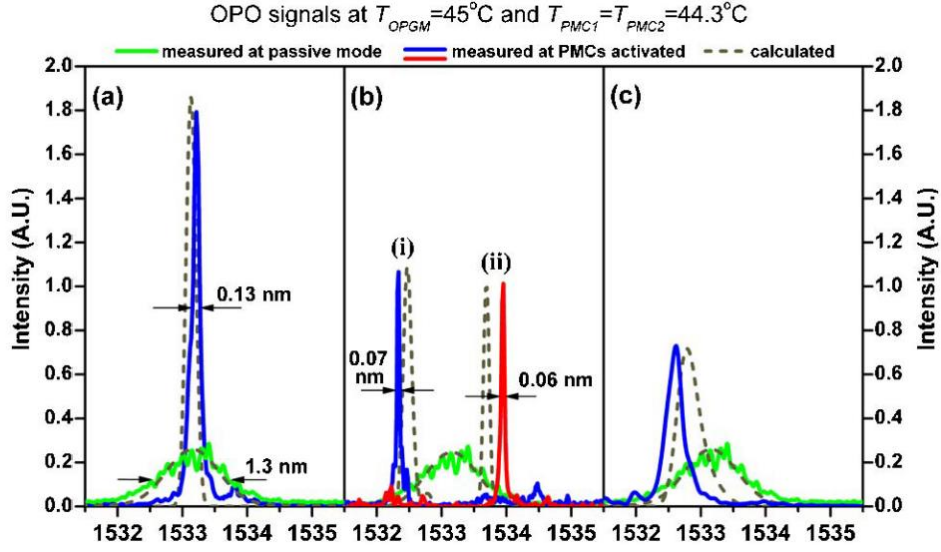


Figure 3-5. (Color on line) Measured (solid black line; blue/red on line) and calculated (dashed gray line) output signal spectra from the OPO system at $T_{OPGM} = 45^\circ\text{C}$, $T_{PMC1} = T_{PMC2} = 44.3^\circ\text{C}$, and at $(E_{PMC1}, E_{PMC2}) =$ (a) (1290, 980) V/mm; (b) (1060, 700) V/mm (i) and (900, 800) V/mm (ii); and (c) (0, 600) V/mm. The solid gray line (green on line) represents the signal spectrum from the system in passive operation.

The vertical scales in Figures 3-5 and 3-6 are directly comparable. The operational parameters used to obtain the presented spectra are summarized in the Figure captions, where T_{OPGM} , T_{PMC1} , and T_{PMC2} denote the operating temperatures of the PPLN OPGM and the first and second PPLN EO PMCs, while E_{PMC1} and E_{PMC2} represent the EO driving fields of the two PMCs. The corresponding OPO signal spectrum from the system with inactive EO PMCs (i.e., with $E_{PMC1} = E_{PMC2} = 0$ V/mm, referred to as the passive mode) is also plotted (solid gray line; green on line) for comparison. We estimated the number of signal transits in the OPO during the signal buildup is ~ 15 according to the linewidths of the passive-mode signals obtained in Figures 3-3 and 3-5[14]. This number of cavity transits has been introduced in our simulation algorithm. In Figure 3-5(a), we obtained a narrowed signal spectral peak from the OPO system with a spectral width (FWHM) narrowed by a factor of ~ 10 and a peak intensity enhanced by a factor of ~ 6 compared to the system in passive

operation. The measured signal peak power is around 6 kW, corresponding to a conversion efficiency of $\sim 26\%$. Fast EO tuning of the narrowed spectral peak was also demonstrated in the presented OPO system. Figure 3-5(b) shows the resultant signal spectra when the EO driving fields of the system are switched to $E_{\text{PMC1}} = 1060$ V/mm and $E_{\text{PMC2}} = 700$ V/mm (noted by (i)) and to $E_{\text{PMC1}} = 900$ V/mm and $E_{\text{PMC2}} = 800$ V/mm (noted by (ii)). We also observed the generation of a narrowed single signal spectral peak in the system with only one of the PMCs activated, as shown in Figure 3-5(c), where the spectral width and peak intensity is obviously less ideal than those obtained in Figures 3-5(a) and 3-5(b).

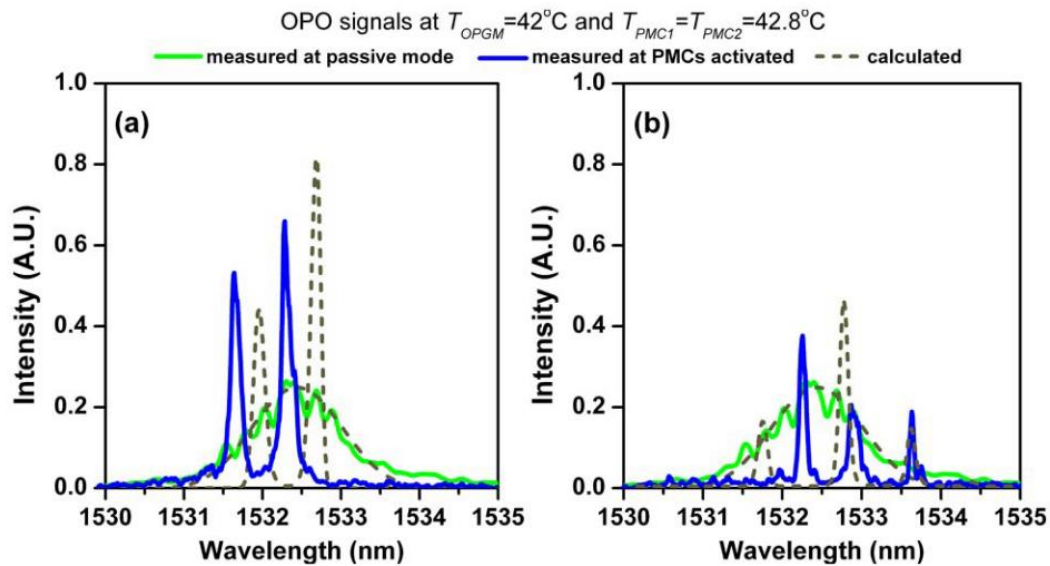


Figure 3-6. (Color on line) Measured (solid black line; blue on line) and calculated (dashed gray line) output signal spectra from the OPO system at $T_{\text{OPGM}} = 42^\circ\text{C}$, $T_{\text{PMC1}} = T_{\text{PMC2}} = 42.8^\circ\text{C}$, and at $(E_{\text{PMC1}}, E_{\text{PMC2}}) =$ (a) (940, 1100) V/mm and (b) (1040, 1050) V/mm. The solid gray line (green on line) represents the signal spectrum from the system in passive operation.

Figure 3-6 shows the results when the system is operated to produce two and three signal spectral peaks in the OPO gain bandwidth with the presented technique. Notably again, the output spectra shown in Figures 3-6(a) and 3-6(b) can be switched instantaneously by changing the applied electric fields using the EO effect. No

well-resolved multiple signal spectral peaks can be observed from the system with only one PMC activated. The discrepancy between measured and calculated spectra in Figures 3-5 and 3-6 may be mainly attributed to the temperature gradients at the interfaces of the PPLN OPGM and EO PMCs sections in the monolithic device configuration.

3.5 Chapter summary

We have successfully demonstrated a 1064 nm pumped OPO system at a 1532 nm band based on a monolithic LiNbO₃ integrating a 3 cm long, 29.8 μm period PPLN OPGM and two 1.25 cm long, 21 μm period PPLN EO PMCs to achieve unique spectral narrowing and manipulation of the output signal. The two PPLN EO PMCs can be operated separately and perform like a pair of active spectral filters whose filtering spectra can be controlled to tailor a specific gain spectrum for the OPO using the EO and/or temperature tunings. From the simulation analysis, we find the shape of the OPO spectra results from the overlapping line shape functions of the OPGM and the e-polarized wave spectra after the first round trip in the device, and the linewidths of the output spectral peaks are dependent on the number of signal transits in the OPO during the signal buildup. With this technique, the OPO system generated output spectra with single to multiple signal peaks whose spectral widths were reduced by a factor of up to 10 with peak intensities increased by a factor of up to 6 compared to the output spectrum in passive mode. Fast EO tuning of the narrowed spectral peak has also been demonstrated in the presented OPO system.

Chapter 4

Electro-optic guided-to-radiation mode conversion in annealed proton-exchanged PPLN waveguides

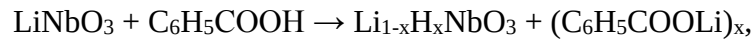
EO control of guided-to-radiation (GTR) mode conversion was first studied by Marcuse in LiNbO₃ (and LiTaO₃) based on the electro-optically induced off-diagonal dielectric perturbation mechanism. With this EO effect in, LiNbO₃ waveguides, the coupling of a TM waveguide mode to a TE radiation mode is readily attainable due to the automatic phase matching of the process where a continuum of radiation modes is involved per the birefringence of LiNbO₃ (a negative uniaxial crystal). However, the modulation efficiency from such a simple modulator was usually very low mainly due to the poor field overlap between the guided mode and the phase-matched radiation mode. In this work, we use the QPM technique to provide a reciprocal vector to compensate the wave-vector mismatch of the condition for the efficient mode-field overlap in the waveguide under the GTR mode interaction process.

4.1 Annealed proton-exchanged LiNbO₃ waveguide

The choice of the APE technique [29] as the waveguide fabrication method in the present work has many advantages. First, APE technique has been one of the most mature approaches to fabricate a stable, low-loss optical waveguide in LiNbO₃; second, APE LiNbO₃ waveguide fabrication is a relatively low-temperature process (<400°C), which is of particular benefit to the QPM waveguide fabrication in contrast with the use of another popular waveguide fabrication method based on titanium diffusion requiring a high processing temperature (>1000°C); and third, only the TM polarized

waves can be guided in a z-cut APE LiNbO₃ waveguide as only the extraordinary refractive index is increased by the proton exchange (in contrast with the Ti:LiNbO₃ waveguides that can support both TM and TE polarization modes). This third feature of APE LiNbO₃ waveguides is important for the accomplishment of the TM-guided to TE-radiation mode conversion.

The fabrication of APE channel waveguides in LiNbO₃ has been a well-established technique [29], and we discuss it briefly in the following. APE LiNbO₃ waveguides can be fabricated by a two-step process. The first step is called proton exchange (PE) process. Such a process is based on the substitution of Li⁺ ions of LiNbO₃ by protons (H⁺ ions) of acids at the LiNbO₃ surface. We use benzoic acid (BA, C₆H₅COOH) in our PE process due to its weak acid and low melting point around 150°C. The reaction of the PE process is described as follows:



where x is the substitution rate of hydrogen. After the PE process, it generates a waveguide with a step-like index profile of a large increase in the extraordinary refractive index ($\Delta n_e \sim 0.1$) at the LiNbO₃ surface. At this time, there exists significant degradation in *electro-optic* and *nonlinear* coefficient of LiNbO₃ surface and the waveguide propagation loss is relatively high (~ 2 dB/cm). Subsequent annealing process (APE) will reduce the hydrogen concentration in the LiNbO₃ surface and will result in a graded index profile and lower refractive index, as shown in Figure 4-1. Besides, this further annealing also recovers the partial electro-optic and nonlinear properties of the LiNbO₃ and reduces the propagation losses to < 0.4 dB/cm with $\Delta n_e \leq 0.03$.

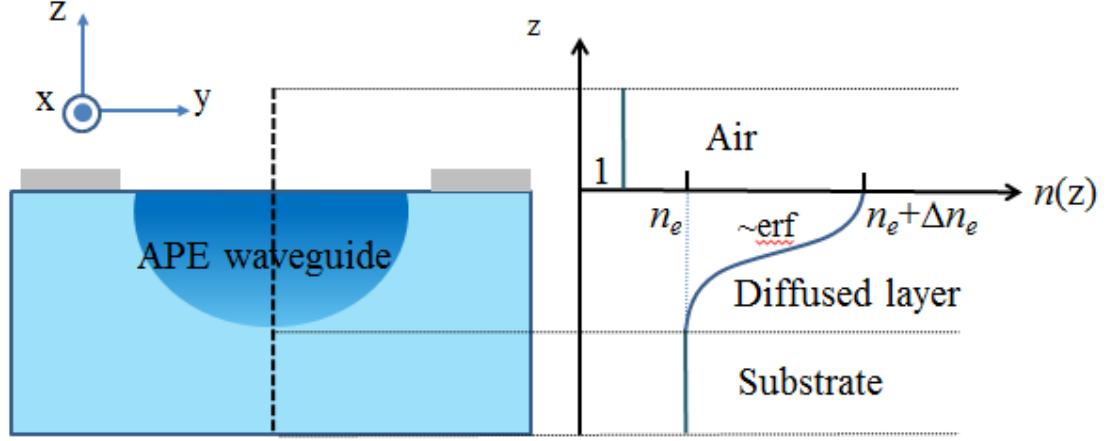


Figure 4-1. Scheme of the extraordinary refractive index distribution $n(z)$ along the central vertical dashed line of our APE waveguide.

4.2 Device design

In this work, we implemented the EO GTR PMC in z-cut annealed proton-exchanged PPLN waveguides along the crystallographic x-axis direction to demonstrate efficient coupling between the TM_{00} mode and TE radiation modes. In the presence of an external electric field E_y along the PPLN crystallographic y axis, the dielectric tensor of the crystal is periodically perturbed along the x (domain grating vector) direction, inducing the efficient energy coupling between two polarization modes. An analytical expression for describing such a coupling process in guided-to-guided (GTG) mode scheme has been established based on coupled mode theory [13]. However, the treatment for a GTR scheme would be more complex since the conversion process is associated with a continuum of radiation modes. The coupled mode equations [30] describing an EO TM_{00} -guided to TE-radiation mode conversion process in a PPLN waveguide can be expressed as:

$$\begin{aligned} \frac{dA_g}{dx} &= \int_0^{n_{os}k_0} \kappa_{gr} A_{\beta_p} e^{-i\Delta\beta x} d\rho \\ \frac{dA_{\beta_p}}{dx} &= \kappa_{gr}^* A_g e^{i\Delta\beta x} \end{aligned} \quad , \quad (4.1)$$

,where the integration in the first equation counts over a continuum range of radiation

modes in codirectional coupling scheme. In Eq. (4.1), A_g and A_{β_ρ} are the slowly varying field amplitude envelopes of the TM-guided and TE-radiation modes, respectively, k_0 is the wave number in vacuum, n_{os} is the ordinary refractive index of the substrate, i is the imaginary unit $\sqrt{-1}$, $\Delta\beta$ is the wave-vector mismatch for the QPM interaction of the APE PPLN waveguides, defined by

$$\Delta\beta = \beta_\rho - \beta_g - K_{\text{QPM}} \quad (4.2)$$

where β_ρ and β_g are the longitudinal propagation constants of the TE-radiation and TM-guided modes, respectively, and $K_{\text{QPM}} \equiv 2\pi m/\Lambda$ is the m^{th} -order grating vector of the PPLN with period Λ , $\rho = \sqrt{n_{os}^2 k_0^2 - \beta_\rho^2}$ represents the transverse wave number of the TE-radiation modes, and κ_{gr} is the coupling coefficient normalized to the power flow carried by the mode, given by

$$\kappa_{gr} = -i \frac{2}{\lambda_0} \frac{E_0}{\sqrt{n_{o,eff} n_{e,eff}}} \frac{\sin(m\pi D)}{m} \mathcal{G} \quad (4.3)$$

where λ_0 is the wavelength in vacuum, E_0 denotes the ideal uniform field amplitude of the applied electric field E_y , $n_{o,eff}$ and $n_{e,eff}$ are the effective refractive indices of the ordinary and extraordinary (corresponding to TE and TM) modes, respectively, D is the PPLN domain duty cycle, and $\mathcal{G}$ is the overlap integral defined by

$$\mathcal{G} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} r_{51}(y, z) (n_o(y, z) n_e(y, z))^2 e_{TE}^*(y, z) e_y(y, z) e_{TM}(y, z) dy dz, \quad (4.4)$$

where $r_{51}(y, z)$ is the relevant Pockels coefficient, $n_o(y, z)$ and $n_e(y, z)$ are the ordinary and extraordinary refractive indices, $e_{TE}(y, z)$ and $e_{TM}(y, z)$ are the normalized transverse field profiles of the TE and TM modes, and $e_y(y, z)$ is the cross-sectional field distribution of E_y in the present planar electrode configuration [17]. Because of the convenience of the power collection of a guide wave, the conversion efficiency of

such a GTR mode converter is typically determined as the fractional power loss of the guided mode in it at the application of the EO field, given by

$$\eta_c = 1 - \left| \frac{A_g(L)|_{E_y \neq 0}}{A_g(L)|_{E_y = 0}} \right| \equiv 1 - T_{EO} \quad (4.5)$$

where L is the length of the converter and T_{EO} is the power transmittance defined as the ratio of the guided-mode power output from the waveguide with a finite E_y to that with $E_y = 0$.

However, an analytical solution of the coupled mode equations given in Eq.(4.1) cannot be found due to the involvement of the integral over ρ ; nevertheless, it can be readily concluded from Eqs. (4.1), (4.3), and (4.4) that the coupling coefficient κ_{gr} , and therefore the overlap integral $\mathcal{G}$ (or termed as “transverse phase matching” [19]), is the crucial parameter that determines the efficiency of the device. According to Eq.(4.2), the PPLN grating period Λ can thus be a useful degree of freedom to phase-match the coupling to a radiation mode having a proper transverse field distribution to maximize $\mathcal{G}$. To find an optimum PPLN grating period for the present EO GTR PMC, we studied the overlap integrals of the mode couplings in this device with the aid of the numerical finite-element method [6] for the mode fields calculation in a diffused waveguide. Figure 4-2 shows the calculated normalized overlap integral $\mathcal{G}$ according to Eq. (4.4) as a function of the effective refractive index $n_{o,eff} (\beta_\rho / k_0)$ and the corresponding escaping angle θ [31] of the radiation mode for the coupling between the TM_{00} and TE radiation modes in an APE LiNbO₃ waveguide at $\lambda_0 = 1520$ nm, at 40°C. The red line represents the fitting curve of the calculated results distribution. The dashed green line indicates the corresponding value of the effective refractive index of the TM_{00} guided mode, which is $n_{e,eff} \sim 2.1489$. It clearly shows the TM_{00} guided mode (automatically) phase-matches to a radiation mode escaping at a large angle ($\sim 14^\circ$), resulting in a much lower overlap integral value. The overlap

integral distribution peaks around an effective refractive index of $n_{o,eff} \sim 2.2120$, approaching to the ordinary refractive index of the substrate $n_{os} = 2.2123$. The radiation mode with this effective index has an escaping angle of $\theta \sim 1.2^\circ$.

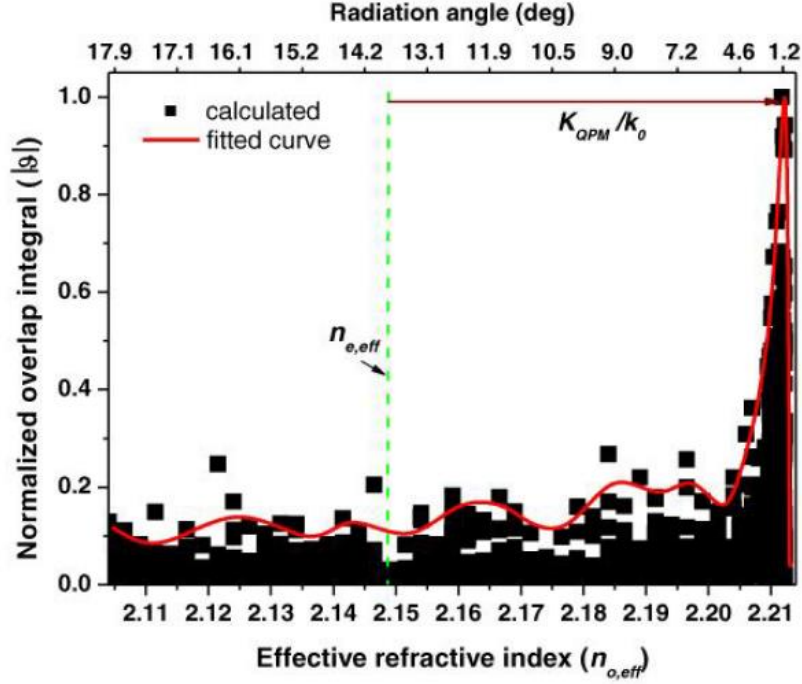


Figure 4-2. Calculated normalized overlap integral as a function of the effective refractive index and the corresponding escaping angle of the radiation mode for the coupling between the TM_{00} and TE radiation modes in an APE LiNbO₃ waveguide at $\lambda_0 = 1520$ nm, at 40°C. The red line is the fitting curve of the calculated results distribution. The dashed green line indicates the corresponding value of the effective refractive index of the TM_{00} guided mode. The arrow marked with K_{QPM}/k_0 means the PPLN grating vector demanded to satisfy the longitudinal phase matching for an efficient overlap integral.

The mode-field overlap efficiency and therefore the GTR mode conversion efficiency can thus be highly enhanced if the conversion process is conducted in a near-collinear scheme (where radiation modes have small transverse wave numbers [19]). The PPLN grating period demanded to satisfy the longitudinal phase matching of this near-collinear conversion scheme can be derived from Eq. (4.2) with $\Delta\beta = 0$, given by

$$\Lambda = \frac{m\lambda_0}{n_{o,eff} - n_{e,eff}}, \quad (4.6)$$

which yields $\Lambda = 24.1 \text{ } \mu\text{m}$ for $m = 1$.

4.3 Device fabrication

We first fabricated five 1.7-cm long, 2-mm wide PPLN grating sections of periods $\Lambda = 10, 24, 30, 50$, and $72 \text{ } \mu\text{m}$ along with a single-domain section in a monolithic LiNbO₃ crystal of 0.5-mm thickness using the standard electric-field poling technique [23]. The $24 \text{ } \mu\text{m}$ is expected to be the 1st-order birefringence QPM grating period for the coupling between the 1520-nm TM₀₀ mode and near-collinear TE radiation modes in an APE PPLN waveguide at around 40°C according to Eq.(4.6). The other domain sections were fabricated for the experimental comparisons. We then fabricated an array of APE channel waveguides, with a center-to-center spacing of $125 \text{ } \mu\text{m}$, in the made PPLN crystal (i.e., there are 16 waveguides in each 2-mm wide domain section). In our design, these waveguides can support single-mode (TM₀₀) operation in a spectral range of interest, 1495-1640 nm. These waveguides had a proton-exchanged (PE) width of $9\text{-}\mu\text{m}$ and were fabricated by immersing the crystal in the molten benzoic acid at 180°C for 6-hour proton exchanging, then followed by an annealing process in air at 350°C for 14 hours ($\Delta n_e \approx 0.03$; the extraordinary change is measured with the prism coupler, together with the IWKB [32] method). The characterized waveguide depth is around $4.1 \text{ } \mu\text{m}$. To apply an electric field along the crystallographic y axis of PPLN over all the waveguides, we fabricated two interleaved comb-shaped electrodes with teeth aligned along the waveguide sides and spaced by $40 \text{ } \mu\text{m}$ on the + z surface of the monolithic crystal. The effective electrode length is 1-cm long. Both end faces (x surfaces) of the crystal were optically polished, but without the application of an optical coating. Figure 4-3 shows a microscopic image of a portion of the + z

surface of the accomplished APE PPLN EO waveguide device. The HF etched -z surface is also visible in the image, revealing the PPLN gratings with a domain duty cycle of ~45%.

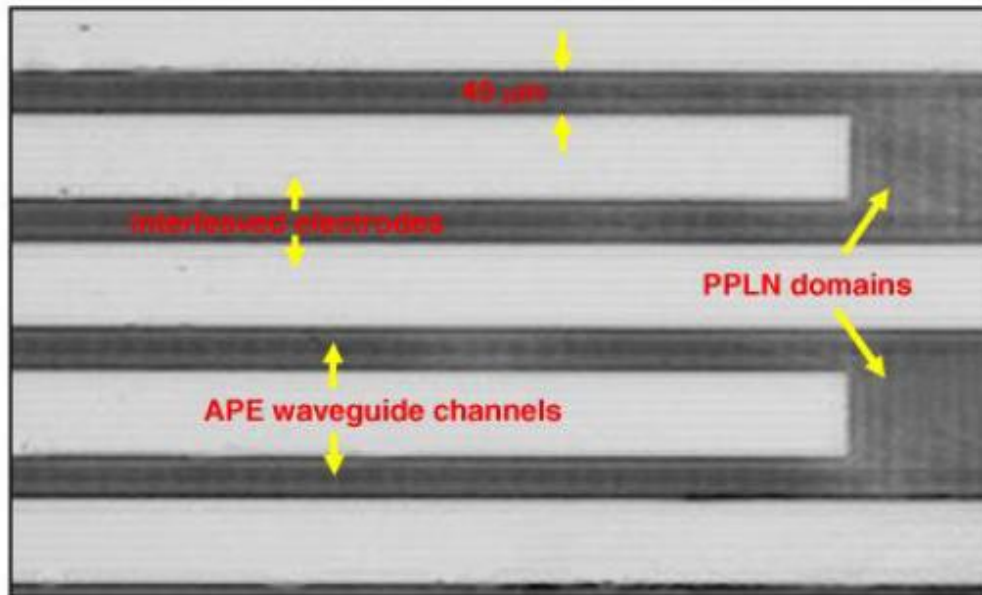


Figure 4-3. Microscopic image of a portion of the + z surface of the fabricated APE PPLN EO waveguide device, showing the arrangement of the electrodes relative to the waveguides and the PPLN domains.

4.4 Performance characterization

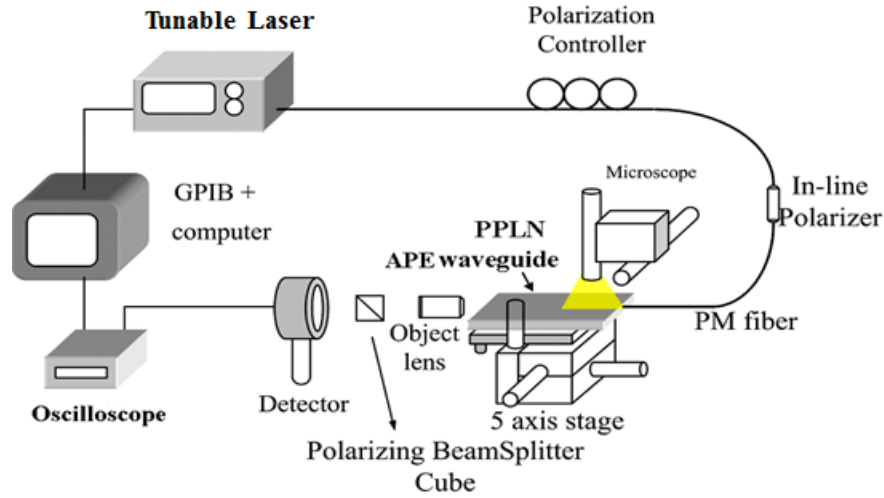


Figure 4-4. Experimental setup for the performance characterization of the APE PPLN EO waveguides.

As shown in Figure 4-4, the performance of the APE PPLN EO waveguide device as a GTR PMC was characterized in an optical testbed with a fiber-coupled external cavity laser (ECL) tunable in a spectral range of 1495-1640 nm as the light source. The output of the ECL was polarization maintained and connected to an in-line polarizer whose input and output were coupled with single-mode, polarization-maintaining (PM) fibers to fix the polarization state of the laser beam along the optic axis (i.e., z axis) of the crystal to fully excite the TM mode in the waveguide using the butt-coupling scheme. The coupled laser power in the waveguide was around 1.1 mW with an estimated fiber-waveguide coupling efficiency of -2.2 dB. The crystal was installed in a temperature controlled oven followed by a polarization beam splitter at the output end as an analyzer. We first measured the propagation loss of the APE PPLN waveguides using the Fabry-Perot method [33] and found it to be around 0.3 dB/cm. Figure 4-5 shows the measured spectral transmittances (T_{EO} ; refer to Eq. (4.5)) of the TM mode from a 24- μ m-period APE PPLN waveguide for various

EO voltages (V_{EO}) at 35°C. The inset shows the measured transmittance of the TM guided mode at 1520 nm as a function of the crystal temperature from the 24- μm -period PPLN waveguide at an EO voltage of 250 V, indicating a moderate dependence of the coupling efficiency on temperature in a PPLN EO GTR mode converter. According to Figure 4-5, a maximum mode conversion efficiency η_c ($\equiv 1 - T_{EO}$) of >95% was obtained at around 1520 nm with 250-V EO voltage. This efficiency has been remarkably enhanced by a factor of >10 over a prior work in a Cu-ion diffused LiTaO₃ waveguide using a periodic-electrode structure under the same interaction length (1 cm) and EO field (~ 6.3 V/ μm) [19]. It is also to the best of our knowledge the most efficient GTR mode converter ever reported based on LiNbO₃ on-axis waveguides.

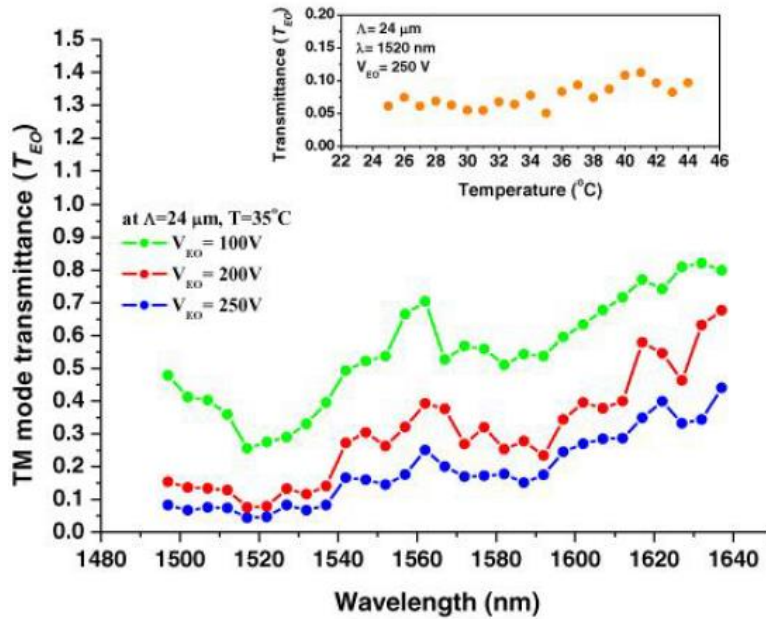


Figure 4-5. Measured spectral transmittances (T_{EO}) of the TM mode from a 24- μm -period APE PPLN waveguide for various EO voltages at 35°C. The inset shows the measured transmittance of the TM guided mode at 1520 nm as a function of the crystal temperature from the 24- μm -period PPLN waveguide at an EO voltage of 250 V.

Besides, a broad conversion bandwidth of ~ 250 nm (at $V_{EO} = 250$ V) was also estimated from Figure 4-5, which is a signature of the GTR mode conversion, in contrast with a typical bandwidth of < 3 nm obtained from a PPLN EO GTG mode converter with the same electrode length [13]. To investigate this phenomenon, we calculated Eq. (4.4) for different wavelength at 40°C as that in Figure 4-2. Because it is time-consuming to obtain Figure 4-2 based on the two dimensional APE waveguide, we calculated Eq. (4.4) based on the planar waveguide for estimation of the broad conversion bandwidth. As shown in Figure 4-6(a), the planar waveguide depth is $4.1\ \mu\text{m}$ and the extraordinary refractive index and change are 2.138 and 0.0289, respectively. Figure 4-6(b) shows the calculated normalized mode overlap integral as a function of the effective refractive index for the coupling between the TM_{00} and TE radiation modes in the planar waveguide at $\lambda_0 = 1520$ nm at 40°C . Figure 4-6(b) is obtained by using the formula of the radiation mode and guided mode distribution in [17]. The effective refractive index of the overlap integral distribution peak is $n_{o,p} \sim 2.21$. The dashed green line indicates the corresponding value of the effective refractive index of the TM_{00} guided mode, which is $n_{g,eff} \sim 2.162$.

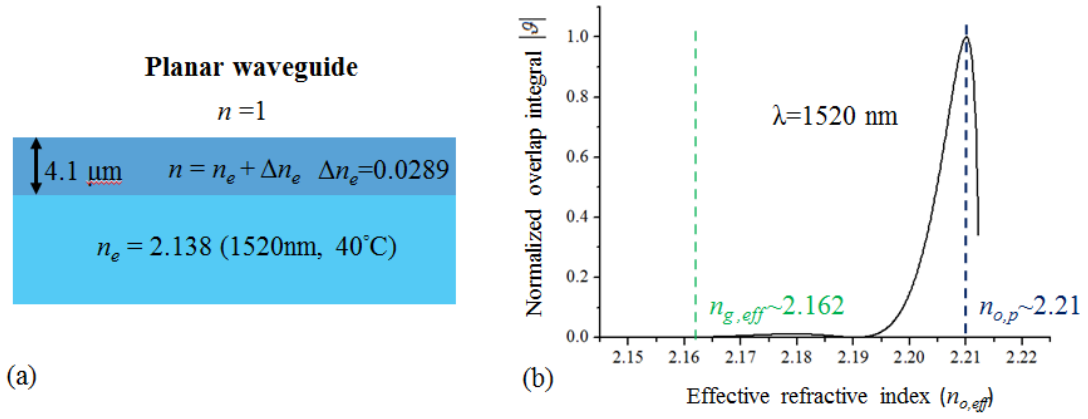


Figure 4-6. (a) is the schematic pattern of the planar waveguide which is used for estimation of the broad conversion bandwidth. (b) shows the calculated normalized overlap integral as a function of the effective refractive index for the coupling between the TM_{00} and TE radiation modes in the planar waveguide at $\lambda_0 = 1520 \text{ nm}$ at 40°C . The dashed green and blue lines indicate values of the effective refractive index of the TM_{00} guided mode and the overlap integral distribution peak, respectively.

The PPLN grating period demanded to satisfy the longitudinal phase matching of these two modes can be derived from Eq. (4.2) with $\Delta\beta = 0$ and is $\Lambda = 31.8 \mu\text{m}$ for $m = 1$. Figure 4-7(a) and 4-7(b) show the mode overlap integral as a function of the effective refractive index for the coupling between the TM_{00} and TE radiation modes in the planar waveguide at $\lambda_0 = 1500 \text{ nm}$ and 1580 nm , respectively, at 40°C . The value of the mode overlap integral in Figure 4-7 is divided by the maximum of the mode overlap integral in Figure 4-6(b) without normalization. By taking the PPLN grating period into consideration, we show the effective refractive indices of phase-matching radiation modes which are denoted by the n_{pm} with respect to the TM_{00} guided modes in Figure 4-7. In Figure 4-7, it shows that the value of the mode overlap integral between the phase-matching radiation mode and the TM_{00} guided mode is appreciable even at wavelength of 1580 nm . It should be the reason that our APE PPLN waveguide with a broad conversion bandwidth.

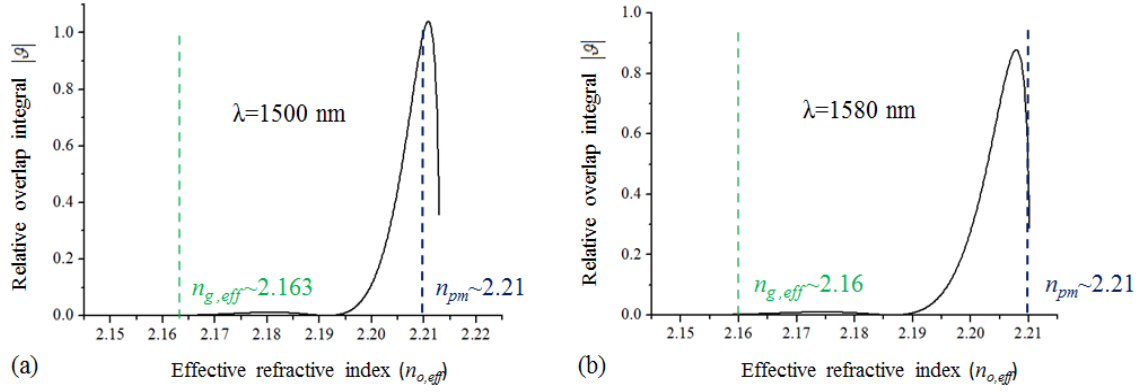


Figure 4-7. It shows the mode overlap integral as a function of the effective refractive index for the coupling between the TM_{00} and TE radiation modes in the planar waveguide at $\lambda_0 = 1500$ nm and 1580 nm at 40°C. The dashed green and blue lines indicate values of the effective refractive index of the TM_{00} guided mode and the phase-matching radiation mode, respectively.

Figure 4-8(a) plots the measured spectral transmittances of the TM mode from the APE PPLN EO waveguides of various grating periods including $\Lambda \rightarrow \infty$ (the period of the single-domain section) when operated at 250 V and 35°C. The experimental results with periods of 10 and 50 μm (not shown) are similar to those obtained with the single-domain one. Figure 4-8 (b) plots the mode conversion efficiency η_c of the device as a function of the PPLN period from the experimental results at 1520-nm wavelength.

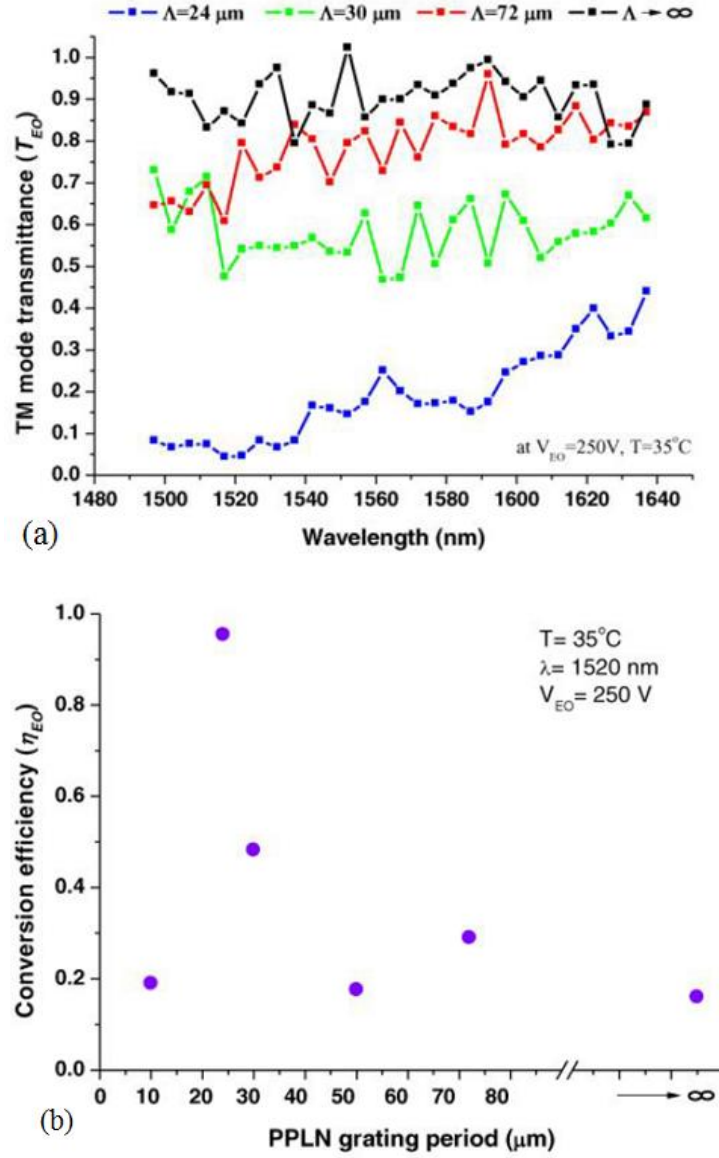


Figure 4-8. (a) Measured spectral transmittances (T_{EO}) of the TM mode from the APE PPLN EO waveguides of various grating periods (not shown for $\Lambda = 10$ and $50 \mu\text{m}$) when operated at 250 V and 35°C . (b) Mode conversion efficiency (η_c) of the device as a function of the PPLN period at 1520-nm wavelength.

As expected, APE PPLN EO waveguides of a period satisfying the birefringence QPM condition (i.e., $24 \mu\text{m}$; refer to Eq. (4.6)) for a near-collinear conversion scheme do show the best GTR mode conversion efficiency over others of different periods and attain an efficiency enhancement of more than 4.6 folds over the waveguides built in the single-domain section. An observable GTR mode conversion was also measured

from the 72- μm -period PPLN EO waveguides (dotted red line) due to the 3rd-order ($m = 3$) QPM grating structure.

As shown in Figure 4-9, we can find the z component of the applied electric field (E_z ; along waveguide depth direction) produced in a surface-electrode configuration [17]. The ripple-like spectral transmission curves and the occurrence of some experimental results with transmittance >1 in Figure 4-8 can be attributed to the slight mode-confinement change in waveguides caused by an unwanted EO effect (accessing the EO coefficient r_{33}) arising from the z component of the applied electric field.

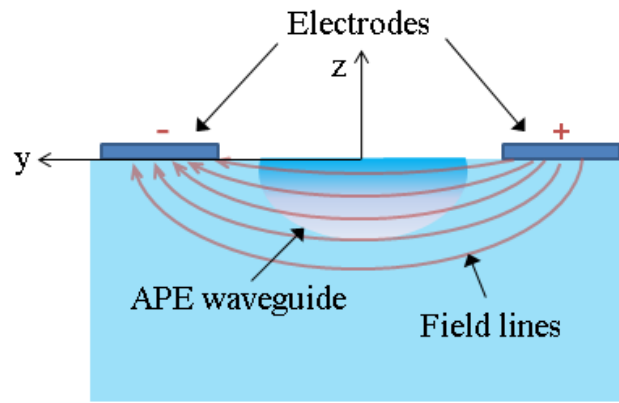


Figure 4-9. Schematic pattern of the applied electric field distribution.

Figure 4-10 shows the far-field intensity profiles of the 1550-nm TM_{00} guided mode measured from an APE waveguide in the single-domain section with an EO voltage of 0V (left) and 250 V (right). Since the EO mode conversion effect induced by E_y around this wavelength is negligible for waves in the single-domain section, the observable discrepancy between the two field-intensity distributions manifests the change of the mode confinement condition due to the presence of E_z .

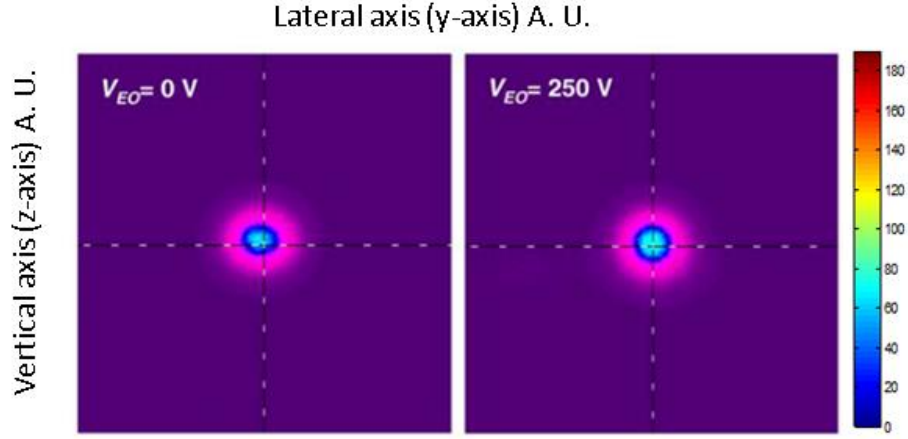


Figure 4-10. Far-field intensity profiles of the 1550-nm TM_{00} guided mode measured from an APE waveguide in the single-domain section with an EO voltage of 0 V (left) and 250 V (right).

Figure 4-11 shows the far-field intensity profiles of the 1520-nm TM_{00} guided mode measured from a 24- μm -period PPLN APE waveguide with an EO voltage of 0V (left) and 250 V (right). The EO mode conversion effect induced by E_y around this wavelength at this waveguide shows the best GTR mode conversion efficiency. Because of the leaky nature of TE radiation modes of such waveguide, the large decrease in the field-intensity distribution at the $V_{EO}=250$ V shown in Figure 4-11 then can be understood.

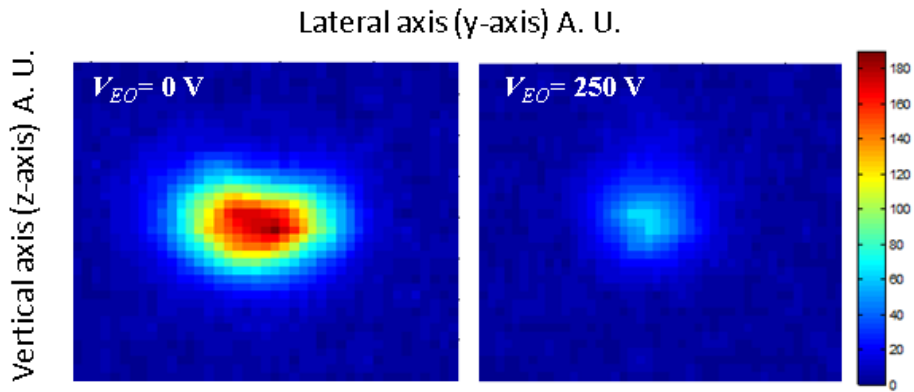


Figure 4-11. Far-field intensity profiles of the 1520-nm TM_{00} guided mode measured from an APE waveguide of grating period 24 μm with an EO voltage of 0 V (left) and 250 V (right).

4.5 Chapter summary

An array of APE channel waveguides were fabricated in a monolithic, multiple grating-period PPLN crystal including a period of 24 μm phase-matching to the 1st-order birefringence QPM coupling between the 1520-nm TM_{00} mode and near-collinear TE radiation modes in the waveguide at 35°C. We achieved a mode conversion efficiency of $>95\%/ \text{cm}$ with an estimated $\sim 250 \text{ nm}$ bandwidth from the 24- μm -period PPLN waveguide at an EO field of $\sim 6.3 \text{ V}/\mu\text{m}$. The obtained efficiency enhancement is more than 4.6 times over a single-domain (unpoled) LiNbO_3 EO waveguide GTR PMC and ~ 2.4 times over a LiNbO_3 off-axis waveguide GTR PMC [34] under the same device length and EO field.

Chapter 5

Conclusion

In this dissertation, we have designed, fabricated, and demonstrated several EO QPM devices by applying the linear EO effect in z-cut PPLNs. When an electric field is applied along the crystallographic z axis of the PPLN, this PPLN then produces an electro-optically controlled Bragg grating. We developed a theoretical model to investigate and analyze the Bragg diffraction behavior of a Gaussian beam interacting with such a device. Furthermore, some unique EO induced effects that are found in the PPLN grating such as the light guiding effect and the FTIR wave coupling process were suggested to be responsible for the formation of a background light distribution and the slight deviation of the diffraction mode position from the model predicted. For the deviation between the measured and calculated dip/peak positions of the 0th order beam, we can add the extra phase shift between the plane wave elements and its 1st harmonic waves of the incident Gaussian beam in the computer code to further examine such phenomenon. However, the background light distribution in the diffracted pattern at the high applied voltage cannot be explained by this model. The theoretical model provided here can be used to a non-Gaussian incident beam profile and general grating profile. Besides, it is easy to deal with the diffraction problem involving with high order harmonic waves in the grating. This model can also be used to evaluate the performance of volume gratings in an optical system.

When an electric field is applied along the crystallographic y axis of the annealed proton-exchanged PPLN waveguides, we report the design and experimental demonstration of electro-optically active TM-guided to TE-radiation mode converters. We achieved a mode conversion efficiency of >95%/cm with an estimated ~250 nm bandwidth from the 24- μm -period PPLN waveguide at an EO field of ~6.3 V/ μm .

Since high-quality optical waveguides have been developed to build in PPLN to work as single-pass, highly efficient optical frequency mixers (OFM) [35], the further integration of the present PPLN EO GTR PMC with a PPLN waveguide OFM could be an interesting direction of future development for various devices potential for, e.g., optical communications of high data transmission capacity.

When the device (mentioned in chapter 3) is working with an electric field along the crystallographic y axis of the PPLN, two induced PPLN EO PMCs were integrated with a PPLN optical parametric gain medium in a monolithic LiNbO₃ to form an OPO system. In this system, we can manipulate the gain spectrum of the OPO via EO and/or temperature control of the EO PMCs. We then obtained multiple output signal spectral peaks of a single one when there is no applied electric field. And, the spectral width of these peaks are reduced by up to 10 times and peak intensity is increased by up to 6 times. Fast EO tuning of the narrowed signal spectral peak has also been demonstrated. However, due to the complexity of the temperature control in this system (ie., three-zone temperature-controlled oven), it is hard to produce the desired steady output signal spectral peaks from the OPO. To overcome such a problem, we may employ aperiodically poled lithium niobate (APPLN) [36] structures to simultaneously phase match two different processes (.,ie EO PMC and OPG) within the same LiNbO₃ crystal in our system. In such a system, we just need one temperature-controlled oven. Since the APPLN essentially integrated an EO PMC and an OPG together, we just need a temperature-controlled oven and a voltage source to produce the multiple output signal spectral peaks, as shown in Figure 5-1.

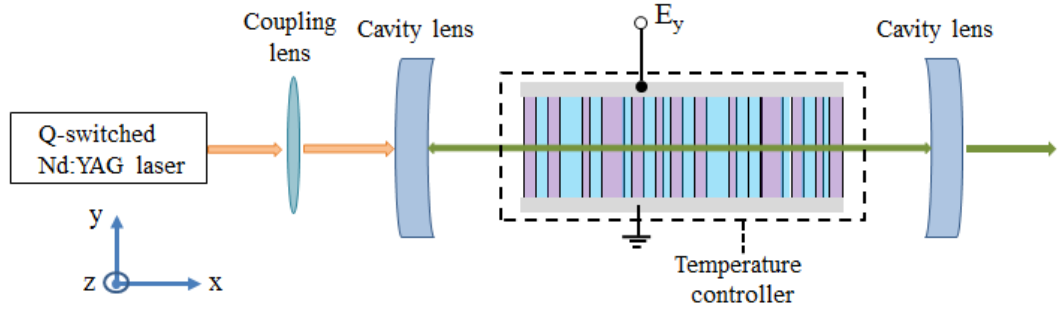


Figure 5-1. Schematic arrangement of the 1064 nm pumped OPO system using an APPLN.

Futhermore, we can integrate the APPLN structure with a PPLN structure as a function of EO Q switch built in a monolithic LiNbO_3 crystal. By employing this device in a diode-pumped Nd:YVO_4 laser, we can achieve an electro-optically tunable intracavity optical parametric oscillator (IOPO) as shown in Figure 5-2.

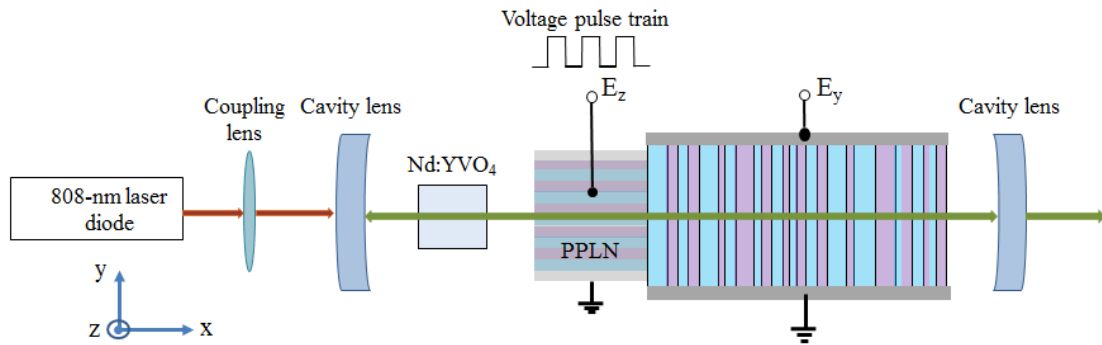


Figure 5-2. Schematic arrangement of the pulsed IOPO constructed in a diode-pumped 1064-nm Nd:YVO_4 laser using the device as simultaneously the laser Q-switch, the electro-optically spectral tuning, and the OPG.

Reference

1. L. E. Myers,* R. C. Eckardt, M. M. Fejer, R. L. Byer, W. R. Bosenberg, and J. W. Pierce ,
“Quasi-phase-matched optical parametric oscillators in bulk periodically poled LiNbO₃, ” J. Opt.
Soc. Am. B. **12**, 2012 (1995)
2. J. A. Armstrong, N. Bloembergen, J. Ducuing, and P. S. Pershan, “Interactions between light waves
in a nonlinear dielectric,” Phys. Rev. **127**, 1918 (1962).
3. M. Yamada, M. Saitoh, and H. Ooki, “Electric-field induced cylindrical lens, switching and
deflection devices composed of the inverted domains in LiNbO₃ crystals,” Appl. Phys. Lett. **69**,
3659 (1996).
4. M. Yamada, “Electrically induced Bragg-diffraction grating composed of periodically inverted
domains in lithium niobate crystals and its application devices,” Rev. Sci. Instrum. **71**, 4010 (2000).
5. Y. Y. Lin, S. T. Lin, G. W. Chang, A. C. Chiang, Y. C. Huang, and Y. H. Chen, “Electro-optic
periodically poled lithium niobate Bragg modulator as a laser Q-switch,” Opt. Lett. **32**, 545 (2007).
6. H. Gnewuch, C. N. Pannell, G. W. Ross, P. G. R. Smith, and H. Geiger, “Nanosecond response of
Bragg deflectors in periodically poled LiNbO₃,” IEEE Photon. Technol. Lett. **10**, 1730 (1998).
7. J. A. Abernethy, C. B. E. Gawith, R. W. Eason, and P. G. R. Smith, “Demonstration and optical
characteristics of electro-optic Bragg modulators in periodically poled lithium niobate in the
near-infrared,” Appl. Phys. Lett. **81**, 2514 (2002).
8. H. Kogelnik, “Coupled wave theory for thick hologram gratings,” Bell Syst. Tech. J. **48**, 2909
(1969).
9. T. K. Gaylord and M. G. Moharam, “Analysis and applications of optical diffraction by gratings,”
Proceedings of the IEEE **73**, 894 (1985).
10. N. Kato, “A theoretical study of pendellosung fringes,” Act. Cryst. **14**, 526 (1961).
11. B. Benlarbi, P. St. J. Russell, and L. Solymar, “Bragg diffraction of finite beams by thick gratings:
two rival theories,” Appl. Phys. B **28**, 63 (1982).

12. Yan-Qing Lu, Zhi-Liang Wan, Quan Wang, Yuan-Xin Xi, and Nai-Ben Ming, "Electro-optic effect of periodically poled optical superlattice LiNbO₃ and its applications," *Appl. Phys. Lett.* **77**, 3719 (2000).
13. C. Y. Huang, C. H. Lin, Y. H. Chen, and Y. C. Huang, "Electro-optic Ti:PPLN waveguide as efficient optical wavelength filter and polarization mode converter," *Opt. Express* **15**, 2548 (2007).
14. S. J. Brosnan and R. L. Byer, "Optical parametric oscillator threshold and linewidth studies," *IEEE J. Quantum Electron.* **15**, 415 (1979).
15. B. Jacobsson, M. Tiihonen, V. Pasiskevicius, and F. Laurell, "Narrowband bulk Bragg grating optical parametric oscillator," *Opt. Lett.* **30**, 2281 (2005).
16. 林昭宏, Integrated Periodically and Aperiodically Poled Lithium Niobate and Photonics and Laser Devices (Doctoral dissertation, National Central University, 2009).
17. D. Marcuse, "Electrooptic coupling between TE and TM modes in anisotropic slabs," *IEEE J. Quantum Electron.* **11**(9), 759 (1975).
18. H. F. Taylor, and A. Yariv, "Guided Wave Optics," *Proc. IEEE* **62**(8), 1044 (1974).
19. Y. Okamura, S. Yamamoto, and T. Makimoto, "Electro-optic guided-to-radiation mode conversion in Cu-diffused LiTaO₃ waveguide with periodic electrodes," *Appl. Phys. Lett.* **32**(3), 161 (1978).
20. 鄧聖龍, Guided-to-radiation polarization mode conversion in electro-optic PPLN APE waveguides (Master's thesis, National Central University, 2008).
21. L. Paul and R. C. Dale, *Electromagnetic Fields and Waves* (W. H. Freeman and Company, 1972), Chap.12.
22. M. Müller, E. Soergel, and K. Buse, "Light deflection from ferroelectric domain structures in congruent lithium tantalate crystals," *Appl. Opt.* **14**, 6344 (2004).
23. L. E. Myers, *Quasi-Phasematched Optical Parametric Oscillators in Bulk Periodically Poled Lithium Niobate* (Ph.D. Dissertation, Stanford University, 1995).

24. M. de Angelis, S. De Nicola, A. Finizio, G. Pierattini, P. Ferraro, S. Grilli, and M. Paturzo, "Evaluation of the internal field in lithium niobate ferroelectric domains by an interferometric method," *Appl. Phys. Lett.* **85**, 2785 (2004).
25. M. Müller, E. Soergel, K. Buse, and M. C. Wengler, "Light deflection from ferroelectric domain boundaries," *Appl. Phys. B* **78**, 367 (2004).
26. J. A. Fleck, Jr. and M. D. Feit, "Beam propagation in uniaxial anisotropic media," *J. Opt. Soc. Am.* **73**, 920 (1983).
27. L. E. Myers, G. D. Miller, R. C. Eckardt, M. M. Fejer, R. L. Byer, and W. R. Bosenberg, "Quasi-phase-matched 1.064- μ m-pumped optical parametric oscillator in bulk periodically poled LiNbO₃" *Opt. Lett.* **20**, 52 (1995).
28. Y. H. Chen and Y. C. Huang, "Actively Q-switched Nd:YVO₄ laser using an electro-optic periodically poled lithium niobate crystal as a laser Q-switch" *Opt. Lett.* **28**, 1460 (2003).
29. M. L. Bortz, and M. M. Fejer, "Annealed proton-exchanged LiNbO₃ waveguides," *Opt. Lett.* **16** (23), 1844–1846 (1991).
30. D. Marcuse, "Coupled-mode theory for anisotropic optical waveguides," *Bell Syst. Tech. J.* **54** (6), 985–995 (1975).
31. N. Mabaya, P. E. Lagasse, and P. Vandenbulcke, "Finite element analysis of optical waveguides," *IEEE Trans. Microw. Theory Tech.* **29** (6), 600–605 (1981).
32. K. S. Chiang, "Construction of refractive-index profiles of planar dielectric waveguides from the distribution of effective indexes," *J. Lightwave Technol.* **LT-3**, 385 (1985).
33. R. Regener, and W. Sohler, "Loss in low-finesse Ti:LiNbO₃ optical waveguide resonators," *Appl. Phys. B* **36** (3), 143–147 (1985).
34. S. Yamamoto, and Y. Okamura, "Guided-radiation mode interaction in off-axis propagation in anisotropic optical waveguides with application to direct-intensity modulators," *J. Appl. Phys.* **50** (4), 2555–2564 (1979).
35. M. H. Chou, J. Hauden, M. A. Arbore, and M. M. Fejer, "1.5 μ m wavelength conversion based on difference-frequency generation in LiNbO₃ waveguides with integrated coupling structures," *Opt.*

Lett. **23**(13), 1004–1006 (1998).

36. B. Y. Gu, B. Z. Dong, Y. Zhang, and G. Z. Yang, “Enhanced harmonic generation in aperiodic optical superlattices,” Appl. Phys. Lett. **75**, 2175 (1999).

